

# Monopsony in Space: Commuting & Labor Market Power \*

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October 31, 2025

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## Abstract

This paper studies the role of commuting costs in shaping labor market power, and the allocation of workers to firms. I build, identify, and estimate a two-sided labor market matching model featuring strategic interactions in wage-setting, commuting costs, and residential choice. Changes to workers' commuting costs affects their allocation to firms, and due to strategic interactions, labor supply elasticity. I study the direct and distributional consequences of a commuting shock through the lens of a subway expansion in Vancouver during the 2010 Winter Olympics. Empirically, workers who gained improved access to the subway network experienced an increase in earnings by 1.5-2% relative to workers with no change in access. I provide suggestive evidence the effect is driven by job switchers who travel farther to new employers. Using the estimated model, I show that the expansion reallocated workers in affected areas to more productive firms. In equilibrium, neighborhoods with improved access experienced an 8% drop in labor market concentration. Simultaneously, workers in other neighborhoods were crowded out from high productivity jobs, and faced worse labor market outcomes. This showcases how improved commuting costs for some neighborhoods has adverse spillover effects on other areas through increased competition amongst workers for high-paying jobs. I further show that 10-15% of the spatial variation in wage markdowns, can be explained by the non-uniform access to firms within a Commuting Zone. This operates through changes in worker to firm sorting, underscoring the role of differential job access in shaping labor market power across space.

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\*I am extremely indebted to Kory Kroft, Ismael Mourifié, Nate Baum-Snow, and Clémentine Van Effenterre for unwavering guidance and support. I also thank Samy Amer, Lady Bolongaita, Michael Boutros, Grant Benjamin, Yanyou Chen, Victor Couture, Stephan Heblich, Mahmood Haddara, Gueorgui Kambourov, Marlène Koffi, Peter Morrow, Sobia Jafry, Stephen Tino, Aloysius Siow, and Román Zárate, for helpful discussions; and seminar participants at University of Toronto SWEAT seminar series, University of Toronto Empirical Microeconomics seminar series, and University of Toronto Urban seminar series. In loving memory, of Jojo Zhang. I acknowledge the use of the Toronto Research Data Center for the completion of this research. All errors are my own.  
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# 1 Introduction

Attractive residential locations need not coincide with desirable employment opportunities. To find good jobs workers need to commute. Commuting, however, can be burdensome, and workers are willing to sacrifice higher pay for a shorter commute (Le Barbanchon et al., 2021; Ahlfeldt et al., 2025). Commuting costs therefore generate imperfect substitutability across differentially located jobs. This gives rise to labor market power, allowing firms to set wages below their competitive level, and affects the allocation of workers to firms (Hotelling, 1929; Salop, 1979).

Understanding the role of commuting costs as a source of labor market power can offer new insights on the welfare, and distributional consequences of policies lowering workers' commuting burden. However, despite the growing empirical evidence on firms' wage-setting power, quantitative urban models assume labor markets remain perfectly competitive, and abstract away from firm heterogeneity (Ahlfeldt et al., 2015; Heblich et al., 2020; Tsivanidis, 2022). On the other hand, a growing body of research documents the importance of heterogeneous preferences in shaping labor market power (Card et al., 2018; Azar et al., 2022; Caldwell et al., 2025). Yet, these heterogeneous preferences are modeled as a bundle, and therefore the role of commuting costs in shaping equilibrium outcomes separately from other idiosyncratic preferences is unclear.

This paper examines the role of commuting costs in shaping labor market power, and the allocation of workers to firms. Lower commuting costs increases the substitutability across differentially located jobs, and incentivizes workers to match with more distant employers. The increase in commutable options reduces labor market power faced by workers. In a general equilibrium setting, I study the direct, and distributional consequences of a shock to commuting costs through the lens of a subway expansion in Vancouver, and use the model to emphasize how changes to commuting costs can have ripple effects across space through increased competition for high-paying jobs.

To empirically assess the impact of changes in commuting costs on workers' labor market outcomes, I leverage the Vancouver subway network expansion in preparation for the 2010 Winter Olympics. For the purpose of connecting the Airport south west of the Fraser river with the downtown area in the north, the Canada Line was added to the SkyTrain network. The Canada Line allowed new neighborhoods access to the SkyTrain network, had significant effects on travel times, and increased the share of transit ridership (Chernoff and Craig, 2022; Herzog, 2025). Leveraging Canadian Employer-Employee data with granular information on workers' residential locations, and station-level coordinates, I use a event study design to show that earnings for workers who gained access to the subway network increased by 1.5% three years after the expansion relative to workers who did not gain access. Worker level panel data allows me to follow workers overtime, alleviating compositional change issues inherent in neighborhood level analyses. I provide suggestive evidence that treated workers were more likely to switch employers after the expansion,

and conditional on switching worked at more distant employers. I also show a higher wage effect for job switchers compared to stayers. Although job switching is endogenous, this exercise offers suggestive evidence that under lower commuting costs workers reallocate to higher paying jobs that are located farther away.

Without further structure, however, it is challenging to isolate the mechanisms underlying the observed empirical results, as it could be driven by a number of factors. First, it is possible that worker productivity changed due to a less stressful commute. Second, firms around the area of expansion may have experienced an increase in labor demand due to increased consumer traffic. Third, the subway expansion may have affected the geographic composition of firms, as well as their entry and exit decisions. Finally, the partial equilibrium empirical design nets out general equilibrium effects, and is therefore silent on the ripple effects of the expansion on other unaffected neighborhoods. This is important when changes in commuting costs for a set of neighborhoods can affect labor market outcomes in other areas through greater competition for jobs, and reduced congestion.

To interpret the empirical findings, while addressing these concerns, and to account for spillover effects across space, I incorporate commuting costs into a general equilibrium model of the labor market which features strategic interactions in wage-setting, residential choice, and rich firm heterogeneity. The model builds on previous work of job differentiation by assuming workers idiosyncratic tastes over non-wage amenities generate upward sloping firm specific labor supply curves ([Card et al., 2018](#); [Lamadon et al., 2022](#); [Chan et al., 2024](#)). Given the spatial distribution of workers, and their disutility from commuting to distant jobs, distance to workplace serves as an observable horizontal amenity that generates match-specific utility, and thus imperfect substitutability amongst identical jobs separately from other unobserved horizontal amenities.

Workers of different observable types choose a neighborhood-employer pair to balance housing costs, and residential amenities, with compensation packages, net of commuting costs. Compensation packages include wages, deterministic non-wage amenities, and stochastic idiosyncratic non-wage amenities. Neighborhoods are characterized by an inelastic housing supply, and exogenous residential amenities. Firms are vertically differentiated by wages and deterministic amenities, and horizontally differentiated by commuting costs and other stochastic preferences. The latter two components generate imperfectly elastic firm-specific labor supply curves. By allowing workers' residential locations to affect workers' labor supply behavior through heterogeneous commuting costs, the model implies a concentration index that varies across neighborhoods. This allows me to assess the distributional effect of shocks to commuting costs on labor market concentration within or across cities.

The proposed labor supply model allows for variation in labor supply elasticity across space. In particular, labor supply elasticity to a firm from a given neighborhood is shaped by workers'

stochastic preferences, that are uncorrelated with residential location, and firm market share in that neighborhood. Due to workers' disutility from commuting firm-neighborhood market shares decline exponentially with commuting distance. As a result, commuting costs generate heterogeneous labor supply elasticities across neighborhoods for any given employer. Specifically, given the independence between stochastic preferences and residential choice, strategic interactions imply that workers living close to their workplace will be on average less sensitive to wage cuts, and more likely to be inframarginal. Changes to workers' commuting costs linearly shifts firms' labor supply curves, and due to the presence of strategic interactions, changes the slope too; affecting average firm labor supply elasticity.

On the demand side, firms with heterogeneous production functions optimally post neighborhood invariant wages that are marked down below marginal product of labor. Assuming firms post uniform wages for workers from all neighborhoods, markdowns are shaped by average labor supply elasticity which is a function of the spatial distribution of the firm's workforce. Since workers geographic distribution shape firms' labor supply, it is possible for some employers to be large in some granular geographic areas, even if in aggregate, or at the commuting zone level, they have negligible labor market shares.

Turning to identification, there are two main blocks of parameters to identify. First, labor supply parameters represented in workers' stochastic preferences over unobservable non-wage amenities, and workers' disutility from commuting. Second, the set of parameters representing firms' heterogeneous production technologies. To identify workers' stochastic preferences over non-wage amenities separate from commuting, I use [Berry \(1994\)](#) quasi-labor supply function at the neighborhood-firm level. The identification challenge stems from the potential correlation between systematic preferences for unobserved amenities, other than commuting, and wages. To overcome this challenge, I follow [Lamadon et al. \(2022\)](#), and [Chan et al. \(2024\)](#) by using an internal instrumental variable approach. The main assumption for identification is that innovations to firm productivity are persistent, while deterministic non-wage amenities are transitory.

To identify workers' commuting disutility, I show that the model implies a gravity equation between workers' place of residence and firms. Although using gravity to identify commuting preferences is standard, unlike other spatial models, traveling to a higher productivity location does not fully determine a workers' deterministic payoff, as longer commutes can be compensated by working for higher paying firms within a given destination. Neighborhood level data is therefore insufficient to identify the semi-elasticity of commuting flows with respect to commuting distance. To address this identification challenge, I control for differences in firms' deterministic utility and leverage within firm variation in labor supply with respect to distance to identify the commute flows elasticity. Combining the commute flows elasticity with the given stochastic preference distribution allows me to back out workers' commuting disutility.

Finally, on the demand side, to identify the elasticity of substitution across workers with different observable types, I use the elasticities implied by the labor supply parameters and invert firms' first-order condition for firm profit maximization. Given the elasticity of substitution, and firm revenue from the balance sheet data, I identify other parameters associated with firms' production technology by inverting the production function under a normalization assumption on within firm productivity differences.

I estimate the model using the Canadian Employer-Employee Dynamics Database (CEEDD). The CEEDD covers the entire population of individuals and businesses with taxable income from 2001 to 2019. I leverage the longitudinal nature of the data to estimate workers' and firms' primitives. I also combine the data with a rental price index at the neighborhood level using the 2016 population Census for Canada, to recover the model implied housing supply. The worker-firm matched data contains balance sheet data, and coordinates on place of residence, and workplace.<sup>1</sup> This allows me to separately identify stochastic preferences from commuting costs. Finally, the available firm balance sheet data allows estimation of the heterogeneous firm production functions.

Estimated parameters reveal an average firm specific labor supply elasticity of 7.5, with average markdown for wages of 84% below marginal product of labor. There is substantial heterogeneity across age groups in particular with respect to their marginal utility of income, and commuting disutility. I find they workers are willing to pay between 1 to 2.7% of annual earnings to reduce their commuting distance by 1 km. There is no gender gap in commuting disutility for younger workers, but the gap widens for older workers, with female workers having a 20% higher willingness to pay for a shorter commute. On the firm side, labor demand estimation reveals significant heterogeneity in firms' total factor productivity, and sharp decrease in returns to firm size, with an average returns to scale of 0.29. Using the estimated firm level productivity parameters, I show that workers with longer commutes sort to higher productivity firms. An increase in commuting distance by 1 km is associated with an increase in firm total factory productivity (TFP) by 1%. The relationship between firm TFP, and commuting is stronger for female workers, who have a higher elasticity by 0.47 percentage points. Showing that female workers require to match with higher productivity firms to when commuting longer distances validates the willingness-to-pay estimates.

Equipped with labor supply and demand primitives, housing supply, and residential amenities, I simulate the model in absence of the Canada Line using 2016 data for the entire economy. This exercise holds constant all exogenous aspects of the economy, such as firm location, firm technology, and within firm worker productivity, as well as travel times on unaffected routes - rules out congestion. Simulation results show that workers in neighborhoods on the Canada Line expe-

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<sup>1</sup>Data includes worker and firm (enterprise) location coordinates. For multi-establishment firms only the location of their corresponding enterprise headquarters is reported. To overcome this issue I impute the location using workers' residential choice. More detail on this in Section 2

rienced an increase in earnings by 1% from the presence of the new subway line, while workers in neighborhoods located on the preexisting network, and non-connected areas, faced a drop in earnings due to the expansion. There was little change in average neighborhood markdowns due to workers' resorting to more productive firms, who offer higher wages, but lower markdowns on average. Workers in neighborhoods with access to the Canada Line resorted to more productive firms, as the average TFP of a given worker's employer in this area increased by 5%. Simultaneously, workers in other areas resorted to less productivity firms. This provides evidence that workers on the Canada Line resorted to higher productivity firms, while crowding out workers from other neighborhoods, underscoring the role of spatial competition for high-paying jobs in shaping the distributional consequences of a transit induced commuting cost shock.

Investigating the effects of the shock on labor market concentration, I find that the average generalized concentration index (GCI) dropped by 8.6% in neighborhoods on the Canada Line, with modest increases in other areas. The effects on concentration are driven by changes in shares on the extensive margin, as well as changes in firm shares within a given neighborhood. Lower commuting costs improves an area's access to firms, while simultaneously reducing the value of the outside option, non-employment. As a result, employment in neighborhoods on the Canada Line increased by 8.4%, and decreased by 0.65% and 0.4% in areas with no subway, and pre-connected areas respectively. This provides evidence that the improved employment accessibility on the Canada Line induced more competition for workers in other area on the extensive margin too. Note that unaffected neighborhoods are larger and have bigger populations, explaining the differences in the magnitude of the estimated effects. In addition, due to the improved accessibility, population on Canada Line neighborhoods increased by 3.75%, resulting in an increase in average rents by 8.5%. The increase in population is driven by changes in accessibility to firms in treated areas.

Overall, simulation results highlight that increased competition for high productivity jobs benefited areas on the Canada Line, but crowded out workers in other areas from high-paying jobs. This is also true on the extensive margin, as employment increased in neighborhoods with access to the Canada Line. These results speak to the distributional consequences of asymmetric commuting cost shocks, corroborating the results in [Manning and Petrongolo \(2017\)](#) and [Chapelle and Ubeda \(2025\)](#) regarding the increased extensive margin competition between workers across space. In addition, it further shows that increased competition also occurs on the intensive margin of firm productivity, as workers on the Canada Line increase their supply to distant, more productive jobs. The effects on markdowns are less clear, as workers in affected areas resorted to higher productivity firms who also have lower markdowns on average. Note that markdowns are needed on average for differences in firm productivity to be translated into differences in worker earnings across neighborhoods.



To further show the effects of asymmetric changes in commuting costs on competition amongst workers for high paying jobs, I simulate the model under the counterfactual scenario of uniform distance between workers and firms. Simulation results show that differential firm access explains 10-15% of the variation in markdowns across space, and 40% of the spatial variation in marginal revenue product of labor. By construction, the variation in both equilibrium objects across space is due to spatial variation in worker to firm sorting. These results speak to the role of heterogeneous commuting costs in shaping the spatial variation in market power.

**Related Literature** My paper contributes to three literatures. First, it relates to the fast-growing literature on local monopsony power ([Manning and Petrongolo, 2017](#); [Heise and Porzio, 2022](#); [Datta, 2023](#); [Pérez et al., 2022](#)). I contribute to this literature by analyzing how asymmetric changes in commuting costs across space affects the spatial distribution of labor market concentration, and worker to firm sorting. [Manning and Petrongolo \(2017\)](#) show that transit shocks increases spatial competition for jobs amongst the unemployed. My results show that such shock increases competition for jobs at higher-paying firms, not only the extensive margin. Conceptually, I extend the role of commuting frictions in [Heise and Porzio \(2022\)](#) to models of job differentiation with rich firm heterogeneity, and granular geographies to capture *local* labor market power via a neighborhood specific concentration index. This is motivated by the empirical evidence on the highly local nature of workers' labor markets ([Manning and Petrongolo, 2017](#); [Monte et al., 2018](#); [Caldwell and Danieli, 2024](#)).<sup>2</sup>

Second, this paper adds to the growing literature on imperfect competition ([Manning, 2003](#); [Staiger et al., 2010](#); [Card et al., 2018](#); [Lamadon et al., 2022](#); [Kroft et al., 2025](#)). By nesting job differentiation into a quantitative spatial model, this paper investigates how an *observable* horizontal amenity, commuting costs, can be a source of labor market power; separately from other *unobservable* preferences. Understanding what shapes workers idiosyncratic preferences is key to studying the sources of labor market power. Recent evidence by [Humlum et al. \(2025\)](#) show that systematic firm wide amenities explain only 14% of the amenity penalty associated with moving to a higher paying job, the rest is attributable to match effects in pay and preferences. Although this evidence stresses the importance of horizontal preferences in shaping labor market power, there remains limited evidence on what characteristics differentiates employers, while simultaneously allowing them to set wages ([Mas, 2025](#); [Kline, 2025](#)). This paper underscores the role of commuting costs in shaping these heterogeneous preferences.<sup>3</sup> Finally, the model implied concentration index at

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<sup>2</sup>In Canada 40% of workers are employed within 5 km of their place of residence ([Statistics Canada, 2006](#)). In the US, 70% of US workers are employed within their county of residence ([Monte et al., 2018](#)). In Germany, 73% of workers work within 5 miles of residential and workplace location ([Caldwell and Danieli, 2024](#)).

<sup>3</sup>Recent work by [Bils et al. \(2025\)](#) argues that workers' unobservable comparative advantage across firms shapes firms' upward sloping labor supply. In my framework, observable commuting costs could be interpreted as an observable characteristic shaping such comparative advantage

a granular location level is closely tied to the outside option index constructed by [Caldwell and Danieli \(2024\)](#), who also allow for commuting distance to shape workers’ employment options. However, they do not allow for firm wage-setting power, a feature of the labor market that distorts the allocation of workers to firms.

Third, my paper contributes to the literature on quantitative urban models ([Ahlfeldt et al., 2015](#); [Monte et al., 2018](#); [Tsivanidis, 2022](#); [Heblich et al., 2020](#); [Zárte, 2022](#); [Baum-Snow and Han, 2024](#)). Conceptually, by relaxing the assumption of perfect competition, and allowing for rich heterogeneity across firms, my framework can be used to study the effect of spatially heterogeneous shocks (e.g. labor demand) on worker firm sorting across space. This generalizes the framework in [Monte et al. \(2018\)](#) who study how spatial shocks affect local employment elasticities, but do not consider the role of labor market power. With regards to identification, leveraging matched worker-firm data with information on residential and workplace locations, this paper offers a new method for identifying commuting disutility under labor market power and firm heterogeneity. In particular the use of within firm variation in labor supply with respect to commuting distance, combined with lagged firm productivity shocks, allows me to separately identify the distribution of stochastic preferences from commuting disutility.<sup>4</sup>

The remainder of the paper is structured as follows. Section 2 describes the Canadian Employer-Employee Dynamic Database. Section 3 provides evidence on the effects of the Vancouver Subway expansion. The model is presented in Section 4. Section 5 discusses identification, estimation procedure, and model estimation results. Section 6 presents model counterfactual simulation results, and Section 7 concludes.

## 2 Data

I use the Canadian Employer-Employee Dynamics Database (CEEDD). The data serves two purposes: First, allows leveraging the Canada Line expansion in Vancouver to provide reduced-form evidence. Second, to estimate the model outlined in Section 4 on the effects of changes in commuting costs, due to subway access, on labor market outcomes. The CEEDD covers the universe of individuals and firms in Canada from 2001 to 2019. The database integrates multiple data sources: One, the T4-slips linked with the Record of Employment (T4ROE) includes data on *annual* earnings from the T4 employment slips (similar to W-2 forms in the US) consisting of any form of employment pay, such as wages, commissions, or bonuses. Employment slips are linked with Record of Employment to form the T4ROE dataset. In a given year, if a worker is linked to multiple employers, due to holding multiple jobs or job switches, I select the job with the highest

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<sup>4</sup>[Le Barbanchon et al. \(2021\)](#) use workers’ stated preferences for commuting and job choice to estimate workers’ willingness to pay for a shorter commute. Instead, this paper implements a revealed preference approach using the universe of workers and firms in Canada.



earnings. Second, the T1 Personal Master File (T1PMF) provides demographic information such as age, gender, and most importantly for this paper, the exact geo-coordinates of workers' residential location (Longitude & Latitude). This information will be used to calculate workers' distance to nearest subway station, and will also be used, as will be described below, to impute *establishment* locations allowing the estimation of the commute flows elasticity with respect to distance to workplace. Finally, the database includes the National Accounts Longitudinal Microdata File (NALMF), which contains firms' balance sheet data and NAICS industry codes. Importantly for this paper, it includes information on firms' revenue, which is crucial for identifying firms' heterogeneous production function parameters. Separate from the CEEDD I use the 25% sample of the 2016 Canadian Population Census at the Census Tract (CT) level to obtain housing costs, which are combined with residential characteristics in a hedonic regression to construct a rental price index for 6300 CTs, in 50 Census Metropolitan Areas (CMAs) & Census Agglomerations (CAs). The third dataset consists of the coordinates of all subway stations forming the Vancouver SkyTrain network, along with their opening dates. Combining this with workers' residential coordinates allows me to compute distance to nearest subway station for all workers in Vancouver in any given year.

I restrict the model estimation and analysis to individuals residing in one of the 50 Census Metropolitan Areas (CMAs) with housing price indices available from the Census. This mainly consists of dropping individuals residing in the territories, and other smaller Census Agglomerations (CAs). Moreover, those outside the age range of 25-60 are also excluded. I also drop individuals with missing gender, age, or residential location geo-coordinates. The CEEDD is derived from tax records, and therefore lacks information on hours worked, hourly wage, or full-time indicators. To address this issue, I follow [Li et al. \(2023\)](#) and further narrow the sample down to full-time equivalent (FTE) workers, defined as those earning at least \$18,000 in 2016 dollars.

On the firm side, firms in the public sector (NAICS 91), education (NAICS 61) or health care sectors (NAICS 62) are dropped from the analysis. I also only keep firms with a total revenue of at least \$50,000 and those with a wage bill lower than 80% of their total revenue. Moreover, I only keep firms who have at least 5 full-time workers as defined by the definition above. Regarding firm location, although I observe the exact geo-coordinates of workers' residential location, firms are defined by their Enterprise ID from the Business Registry, which means only headquarters' location information are available. This is an issue for firms with multiple establishments. Even though they represent around 10% of the set of enterprises, they employ more than 40% of full-time workers. To work around this issue I utilize the joint information on worker-firm matches over the years with the following assumptions. First, I assume each firm has only one establishment per CMA, and second, each establishment has at least 5 full-time workers. Using workers' residential CMA I define an establishment as the Enterprise ID  $\times$  CMA pair. If an establishment has less

than 5 full-time workers I assign its workers to the closest establishment, defined by the CMA of their co-workers. Given that establishments are defined by Enterprise ID  $\times$  CMA pairs, I use workers' residential location geo-coordinates to impute establishment location as the *centroid* of all its workers at any point in time. I then assign the firm to the nearest neighborhood given its imputed longitude and latitude.<sup>5</sup>

To check the validity of this imputation I compute the deviation (euclidean distance) between imputed and actual location for single establishment firms for which their precise location is reported in NALMF. The mean deviation is approximately 23 km, however this distribution is highly skewed with a median of 3 km only. One reason for this issue is that some enterprises, including single establishments, often report their auditor's city even when having all of their workers in another. To further validate the establishment location imputation, I compare the derived commuting to workplace distribution with that of the 2006 Canadian Census. The bottom row of Table A-1 presents the commuting distance to workplace in Canada for the 2006 population census. Comparing these figures with the commuting distribution in Table 1 shows that the derived commuting distances in my data replicate that of the census quite closely. For example, Table 1 shows that 21% of workers commute between 5 and 10 km, whereas this value is 23% in A-1. Panel B of Table 1 presents summary statistics at the *enterprise* level. It shows that the average number of workers within a given enterprise is around 60 and 10% of enterprises are multi establishment units. Given worker and firm locations, I define distance traveled to work as the euclidean distance between workers' residential location and establishment imputed location. I keep worker-establishment pairs with at most 75 km, which consists of more than 95% of matches in any given year.

Table 1 presents summary statistics at the worker and enterprise level for the constructed sample for the period of 2001-2019. All dollar values are in 2016 dollars. Workers earn on average \$ 67,000. There is a gender gap in earnings of approximately 32%, note that I drop workers in the public and health sector, in which women are overrepresented. Around 40% of workers reside in Ontario, the largest Canadian province, 25% in Quebec, and 10% in British Columbia. Focusing on commuting patterns, approximately 38% of workers have a workplace that is within 5 km only of their place of residence. The high share of workers who work close to their place of residence is consistent with the findings of Caldwell and Danieli (2024) who show that 73% of German workers had a workplace within 5 miles of their place of residence, pointing towards the localized nature of the labor market.

Finally, I define a *neighborhood* to be a set of clustered Census Tracts (CTs) within a CMA. I group 6200 CTs into 1034 neighborhoods across Canada. Clustering happens at the CMA level, such that each neighborhood is nested within a CMA. I drop clusters where the minimum distance

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<sup>5</sup>If firms or establishments change locations, the new location is assumed to be a new establishment. This paper abstracts from firm location and the firm problem is always static.

between neighborhood centroid, and any CT in the cluster is greater than 20 km. This ensures the removal of remote areas with a sparse set of neighborhoods. This aggregation allows me to estimate, and compute the model on a granular number of neighborhoods with sufficient variation in distance between origin-destination pairs. Descriptives at the neighborhood level are reported in Panel C of Table 1.

I combine the CEEDD with station level data in Vancouver to study the effects of a subway expansion on labor market outcomes. Station level data is sourced from [citylines API](#). I retrieve stations' longitude and latitude, and opening year. The station level data is then linked to its corresponding CT in the CEED. I discuss sample and data construction for this empirical exercise in Section 3. Importantly, the CEEDD database contains granular residential location data. Combining this with station data allows me to construct annual changes in distance to nearest station for all workers.

## 3 Effect of Subway Access on Labor Market Outcomes

### 3.1 Vancouver SkyTrain Expansion

I first present reduced-form evidence on the effect of reduced commuting costs, as a result of gaining access to a subway station, on workers' earnings and travel distance to work. The empirical analysis will focus on the Metro Vancouver area to leverage the expansion of the SkyTrain transit network in 2009 into a new set of neighborhoods. The Vancouver SkyTrain is the longest automated rapid transit system in the world, it includes three lines connecting downtown Vancouver to suburbs such as Surrey, Burnaby, Richmond and Coquitlam. Prior to 2009 the SkyTrain had only two lines; The Expo Line, built in 1986, and later extended in 1990 and 1994, and runs approximately 22 km through Vancouver, Burnaby and New Westminster; And the Millennium Line, for which operations began in 2002, and spans 25 km from west Vancouver to the skirts of Coquitlam in the east, before looping back again to the downtown area. As illustrated by Figure 1, prior to the Canada line construction there was no rapid rail connecting Richmond in the south-west with downtown Vancouver in the north by crossing the Fraser River. Further, prior to 2016 there was no rapid rail connecting Coquitlam in the north east with the downtown area.

In preparation for the 2010 Winter Olympics, the Canada Line became a priority to connect the airport with the downtown area. As illustrated in Figure 1, the expansion entailed the construction of a new line connecting downtown Vancouver in the North, with Richmond in the south across the Fraser river. Plans for the new line were finalized in 2003, construction started in 2005, and the line was operational ahead of schedule in 2009. This new line runs approximately 20 km with 16 stations from Vancouver International Airport to Waterfront Station. As of 2010, the line

served approximately 100,000 riders on a daily basis (Sroka, 2021).<sup>6</sup> Although both Coquitlam and Richmond were in need of a subway connection with the downtown area, funds and efforts were directed towards the Canada line to host the Winter Olympics. This resulted in plans to extend the Millenium line eastward into Coquitlam to be postponed, and did not start until 2016.<sup>7</sup>

The Canada Line expansion serves as an ideal quasi-exogenous shock to workers' travel costs for several reasons. One, the goal of the expansion was to better prepare Vancouver to host the 2010 Winter Olympics. Therefore one of the main goals of the line was to connect the downtown area with the airport, and not targeted at a specific neighborhood. I leverage this institutional feature, along with panel data on residents of the Metro Vancouver area to identify the effect of gaining access to the subway network on earnings and distance to work. Treated workers will be those in the Richmond area, and control units will span the eastern parts of the Metro area to include residents of Coquitlam, who were waiting for the Millenium line extensions, and sections of Surrey across it from the Fraser river. The individual panel data allows me to rule out residential sorting and other compositional changes that could potentially lead to differences in earnings and work behavior for treatment and control neighborhoods as is usually a threat when assessing the effects of subway expansion using spatial panel data only. Individual-level panel data also allows me to follow individuals overtime independently of changes in place of residence. Two, the expansion had a considerable impact on travel time. Prior to the Canada Line public transit users travelling between Richmond and downtown Vancouver had to rely on the 98 *B-Line bus*. During rush hour the trip could take between 50-60 minutes (Sroka, 2021; Herzog, 2025). The improvement in public transit travel time is reflected in mode choice for workers residing in Richmond, the Census Sub-division most affected by the the new line. According to Statistics Canada Population Census; In 2006 only 11% of workers in Richmond used public transit as their primary mode of travel to work. In 2011, after the introduction of the Canada Line, approximately 20% did so. As illustrated in Figure A-2 this represented a steeper increase in public transit use compared to the rest of the Metro Vancouver area. Although not causal, this provides suggestive evidence that the new line reduced travel time for public transit users to the extent of changing the travel mode distribution. This is indeed supported by the causal evidence found by Herzog (2025), who shows that the Canada Line *causally* increased transit use by 4.6 percentage points, and the increase is due to reduced driving to work. Further, using 2016 General Transit Feed Specification (GTFS) file Herzog (2025) simulates the time savings as a result of the new Canada Line by replacing the new rail line with an express bus, to compute historical travel time. Simulations show that the Canada line reduced travel time by almost 7-10 minutes as a lower bound for any route that

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<sup>6</sup>More details on the SkyTrain network and ridership available [here](#)

<sup>7</sup>[This article](#) discusses the political and municipal landscape prior to this extension, and importantly shows that the Millenium line extension was delayed as a result of the Canada line in preparation for the Winter Olympics.

passes along the Canada Line corridor, representing a decrease in travel time of approximately 15-25% across different times of the day. These reductions in travel time will be used to simulate the counterfactual model in absence of the Canada line to quantify and zoom in on the role of how gaining better access to firms mediates the observed treatment effects.

### 3.2 Empirical Strategy

I leverage SkyTrain station level data which includes station geo-coordinates and opening year to combine with workers' residential locations in the CEEDD to calculate individual-year level distance to nearest subway station. The exact residential and station coordinates allows precise measurement of changes in distance to subway station at a granular geographic level. I use these coordinates to calculate the percentage change in distance to nearest subway station during the 2009 expansion. As illustrated by Figure 2, residents of the Richmond area in the south experienced the largest drop in distance to nearest subway station - *treatment exposure*- after the construction of the Canada line. On the other hand, residents of eastern Surrey and Coquitlam did not gain access to a subway station until the end of 2016 when the Millennium line was extended into Coquitlam. Table 2 shows that the average change in distance to nearest subway station was around 70% conditional on experiencing a non-zero change in subway access.

The goal is to estimate the *level* effect of gaining subway access on earnings and distance to workplace. As such the estimand of interest is a weighted average of the Average Treatment Effect on the Treated (ATT), where weights are assigned across the treatment exposure distribution. Following Callaway et al. (2024) the standard Parallel Trends assumption is sufficient to estimate this *level* effect of gaining subway access. I define treated workers as those with a non-zero change in distance to their nearest subway station between 2008 and 2009, set of workers identified as gaining access to a subway network, to estimate the ATT given below.

$$ATT^{\text{loc}} := \mathbb{E}[\Delta y | \% \Delta dist < 0] - \mathbb{E}[\Delta y | \% \Delta dist = 0] \quad (1)$$

Although possible to run a two-way fixed effects model with a continuous exposure measure, Callaway et al. (2024) show that such specification identifies a weighted average of treatment effects across exposure groups, but suffers from having negative weights and does not integrate up to one. To address this issue, I identify the above ATT by defining treatment  $T_i$  equal to one if a worker experienced *some* drop in distance to nearest subway station, and zero otherwise. Another benefit of discretizing treatment into two groups is that it offers a simple and transparent pre-trends test (Sun and Shapiro, 2022). With a continuous treatment definition a stronger pre-trends assumption is needed, to rule out selection into treatment intensity. However, our goal is to estimate the effect of the *level* effect of gaining access to a subway, not the marginal increase in access. I

also present results using percentage change quintiles as treatment bins to test whether workers who experienced a sharper change in distance experienced a greater treatment effect. Results from this specification present estimates for a weighted ATT within each quintile.

I limit the analysis to workers who's nearest subway station was more than 3 km away in 2008, this rules out comparisons between workers who reside close to the Millenium line (earlier treated) which started operating in 2002, and workers treated later due to the Canada Line construction. This is illustrated in Figure (2) where the gray area represents neighborhoods not included in either treatment or control groups. The time period of interest is 2005-2015 during which no other subway expansion took place, however this does not rule out changes in travel costs due to time variation in congestion, changes in bus routes, or other infrastructure projects. Note that the Expo Line was expanded in 2016 to connect Burnaby with Coquitlam. This extension meant that a part of the control group eventually gained access to the SkyTrain network. This extension was planned since the 1990s, but was set aside for political reasons and to redirect funds for the Canada line in preparation for the Olympics. Restricting the time period of interest to 2005-2015 to be prior to the Millenium line extension.

**Descriptives** Figure 2 illustrates the spatial distribution of treated and control neighborhoods at the Census Tract (CT) level. Comparing this map with Figure 1 shows that areas with the highest treatment intensity are those lying on the Canada line route. Note that neighborhoods on the Expo and Millenium lines are exluded from the sample. Table 2 presents summary statistics at the worker-year level by treatment status before and after the expansion for the estimation sample. Panel A shows that prior to the expansion treated wokrrers had higher average earnings- 78,000 relative to 68,000 for their control counterparts- are closer to the downtown area, are more likely to work in professional industries, and less likely to be in manufacturing or construction. They are however employed at firms with lower total wage bills. The *level* differences between both groups does not preclude labor market outcomes from being on parallel trends prior to the expansion. This will be tested in the dynamic event study specifications below, with  $t = -1$  normalized to 0, with standard error clustered at the individual level.

To estimate the ATT in equation (1) I run the following dynamic event study specification:

$$Y_{igt} = \alpha_i + \alpha_{gt} + \sum_{\tau=-4, \tau \neq -1}^{\tau=6} \beta_{\tau} \mathbb{1}\{\tau = t - 2009\} T_i + \varepsilon_{igt} \quad (2)$$

Where  $Y_{igt}$  is outcome for individual  $i$  in demographic group  $g$ , at time  $t$ . I include individual fixed effects  $\alpha_i$ , and demographic group by calendar year fixed effects given by  $\alpha_{gt}$ . Demographic groups are defined as the interaction of gender and age bins. There are two age bins, 25-40, and 40-59. The dummy variable  $T_i$  takes value 0 or 1 depending on treatment status as outlined above.



The parameters of interest are  $\beta_\tau$  which capture the treatment effect relative to time of Canada Line construction.

Worker level panel data allows me to rule out sorting of workers to neighborhoods with better subway access as the main mechanism. Although residential sorting will respond to changes in the transit network, following *initially* affected workers allows me to rule it out. This is similar to [Pérez et al. \(2022\)](#) who study how the expansion of the Santiago transit system affected workers in closer proximity to the new stations.

### 3.3 Results

Figure 3 presents the estimated coefficients from equation (2). The results indicate an increase in earnings by approximately 1.5% three years following the expansion. This effect is persistent and remains at almost 2% until 2015, 6 years after the expansion. The estimated coefficients prior to the expansion indicate that the treatment and control groups earnings were on parallel trends, lending validity to the empirical design and choice of treatment definition. Specifically, it provides evidence that the Canada Line expansion was not correlated with prior trends in outcomes in treatment and control areas relatively. Additionally, Figure A-3 presents the effects of going from the bottom to top quartile from a TWFE specification under a continuous treatment definition. The estimated coefficients show that an increase in the absolute value of the reduction in distance to nearest subway from the lowest to highest quartile increases earnings by 0.6% as a result of the SkyTrain expansion. Although, this specification leverages variation in proximity to new subway station, offering richer variation, it suffers from the negative weights problem as detailed by [Callaway et al. \(2024\)](#). Overall, the observed effect on earnings could be driven by a variety of mechanisms. First, improved lower commuting costs may have incentivized workers to move to more productive farther away firms that were initially very costly to commute to. It is also possible that stayers benefited from increased productivity due to the less burdensome commute. Finally, it is possible that firms on route of the new line experienced a demand shift, especially those in services. It challenging to empirically isolate the role of each channel. My model-based simulations will allow me to isolate the effect of improved access to firms on changes in workers' labor market concentration, and reallocation channels, while holdin all other productivity related variables constant.

Given these empirical challenges, I provide some suggestive evidence that an important mechanism through which workers benefited from this expansion is the ability to move to more productive employers located farther away. Figure 4 shows that treated workers are 4 percentage points more likely to change their 2008 employer 3 years after the expansion. This represents approximately a 10% increase in the probability of switching employers relative to the control group. Providing some suggestive evidence that switching employers is a primary channel through which the effect

on wages is manifested. To better illustrate this, I split the estimation by job movers and stayers. Movers are defined as workers who changed their 2008 primary employer at any point between 2009 and 2010. Figure 5 illustrates results from estimating equation 2 separately by movers and stayers. Estimated coefficients show that movers experienced a larger gain in earnings due to the new line, reaching 2.5% 3 years after the expansion. Confidence intervals point to statistically significant differences in earning gains between movers and stayers. Figure ?? shows estimated coefficients under a continuous treatment definition, split by movers and stayers. For movers, going from the bottom to top quartile of treatment is associated with just under a 1% increase in earnings three years after the expansion. For stayers this value is cut in half. The difference between both groups is statistically significant. It is important however to note that job switching is endogenous, and splitting the sample by this margin can lead to spurious conclusions. For example, if workers at firms hit by a contemporaneous negative productivity shock were more likely to leave, then this would incorrectly attribute the higher wage gains for job switchers to the subway expansion. To provide more suggestive evidence that the subway expansion alleviated affected workers' commuting burden, and allowed them to move to farther employers, I show that job switchers, defined as those who switched employers between 2009 and 2010, had farther away employers compared to job switchers in the control group. Figure 7 presents estimated coefficients from regressing distance to workplace on time and treatment dummies for the sample of job switchers. Estimates show that job switchers in the treated group worked farther away than their control counterparts, who also switched employers between 2008 and 2010. This corroborates the hypothesis that treated workers faced lower commuting costs, allowing them to travel farther away conditional on switching employers.

Time savings due shorter commuting time is another margin through which workers who gain access to a subway station might benefit. For example, with less time commuting workers are able to work for longer hours, or might be more productive for the same number of hours. To gauge the importance of this mechanism, I estimate treatment effects by treatment intensity quantile, where the lowest quantile are the set of treated workers who experienced the highest change in distance to nearest subway station (e.g new station opened one block away). The rationale is that if time savings are the main drivers of the results in Figure 3 then those living closer to a new subway station should experience a higher gain in their earnings. This is possible to disentangle due to accurate station level coordinate data. Figure 8 presents the results from a static Difference-in-Difference with treatment intensity by quantile. The treatment effect for workers in the first quantile of treatment intensity (group with highest change in distance to nearest station) of 4%, and declines gradually reaching 3% for the third to fifth quantiles. This suggests that time savings from shorter commute times plays a role in explaining the effect on earnings in Figure 3. However, the treatment effect on earnings does not vary across treated workers in the second to fifth quantiles

of treatment intensity, and is relatively stable at 2%. These results suggest that time savings due to shorter commute time plays a non-negligible role in explaining the effect of subway access on earnings, however it is not sufficient to explain the effect on earnings for those who had a lower change in access to subway, nor to explain why amongst those workers there is no evidence that workers closer to the new subway station had a higher treatment effect.

The results presented from this empirical exercise are comparable to the findings by [Pérez et al. \(2022\)](#) who show that the expansion of Santiago’s Subway system increased monthly earnings for workers closer to the new subway stations by more than 1% four years after the expansion. Moreover, the results are qualitatively in line with the findings of [Caldwell and Danieli \(2024\)](#) who show that the introduction of a new high speed rail connecting the small city of Montabaur in Germany, with its relatively larger neighbors, Cologne and Frankfurt, increased workers’ outside options due to the reduction on travel time. They do not, however, examine the effect on wages as a result of this shock. This is in line with the job-level specification suggesting that changes in outside options affected incumbent workers without the need to commute further away for a better match. The effect on earnings is also consistent with the findings of [Bütikofer et al. \(2024\)](#) who show that Swedish workers living in close proximity to a new bridge (Öresund bridge) connecting Malmo with Copenhagen experienced an increase in wages by 13% eight years after the opening of the bridge. The larger magnitude of their estimates points to the large difference between the two cities. A difference that is likely lower within the neighborhoods of the Metro Vancouver area. Nevertheless the results presented here is consistent with their findings; better market access increases workers’ earnings.

The empirical evidence suggests that gaining access to the subway network positively affected workers’ earnings. The transit shock has several implications. First, via reducing commuting costs it increases workers’ effective number of employers, this jointly affects their labor supply elasticities, and their allocation to firms. Second, it is possible that treatment effects are explained by other productivity related explanations, such as improved worker productivity due to having a less burdensome commute. Finally, another concern is that untreated workers could potentially benefit from reduced congestion as a result of the subway expansion. This would violate the SUTVA assumption, indicating that spillovers could be a threat to identification of treatment effects. To address these empirical concerns, and to isolate the role of improved access to more productive firms from other explanations, I build on [Chan et al. \(2024\)](#) to identify a two-sided labor market matching model with firm wage-setting power, and commuting costs. The model nests a new classical monopsony framework into a quantitative spatial model, providing a tractable framework to assess the effect of changes in commuting costs on worker firm matching and changes in labor supply elasticity.

## 4 Model

### 4.1 Environment

**Setup** At each point in time  $t$ , there is a mass of  $M_t$  workers  $i \in \mathcal{I}$  who are heterogeneous in their type  $k \in \mathcal{K} := \{1, \dots, K\}$ . Worker type is exogenous, and can be thought of as a set of observable demographics. There is a finite number of cities  $c \in \mathcal{C} := \{1, \dots, C\}$ , where each city  $c \in \mathcal{C}$  has a finite set of neighborhoods denoted by  $z \in \mathcal{Z}(c)$ . Each neighborhood  $z$  has an *exogenous supply of housing* denoted by  $H_{zt}^s$ . Each worker type  $k$  has workers with mass  $m_{kt}$  such that  $\sum_{\mathcal{K}} m_{kt} = M_t$  in each period. On the other hand, there is a finite number of heterogeneous firms  $j \in \mathcal{J} := \{1 \dots J\}$ . Firms are *exogenously* distributed across the finite neighborhoods  $\mathcal{Z} := \bigcup_c \mathcal{Z}(c)$ .<sup>8</sup> Each firm belongs to some sector  $g \in \mathcal{G} := \{1 \dots G\}$  defined by the 2-digit NAICS code of the firm. Firms produce a set of perfectly substitutable freely tradable goods using different types of labor  $l_k$  as the only input into the production function  $F_j(l_j)$ .<sup>9</sup>

**Decisions** Workers take their type  $k$  as given, and choose which neighborhood  $z$  to reside in, and which employer  $j$  to work for. This decision takes into account that attractive residential locations need not overlap with productive ones. Thus, workers make choices to balance housing costs, wages and commute distance.<sup>10</sup> For example, a worker with a long commute is compensated for such disamenity either by higher wages and/or lower housing costs. Firms on the other hand, take their productivity, amenities, and location  $o(j)$  as given, and make hiring decisions by setting wages for each type of worker  $k$ . Firms are vertically differentiated by wages and systematic amenities, and horizontally differentiated due to worker specific variation in distance to workplace, and other idiosyncratic preferences. Conditional on worker type,  $k$ , the horizontal differentiation components induce an upward sloping firm-specific labor supply curve. Firm specific labor augmenting TFPs depend on worker type,  $k$ , but are invariant to workers' locations. Therefore, firms only account for type specific aggregate labor supply. Location of workforce matters to the extent of changes in the marginal worker labor supply elasticity. *The implicit assumption here is that employers can only observe average labor supply curve for a given group  $k$ , and therefore cannot discriminate by worker residential neighborhood,  $z$ .* When making hiring decisions firms decide on how a change

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<sup>8</sup>This is equivalent to allowing firms to choose their location but are assumed to be ex-ante homogenous, and only observe their productivity following location decision. Some recent papers allow for worker-firm sorting across space within a wage setting framework but assume that firms are either atomistic or ex-post homogenous (Bamford, 2021; Hong, 2024; Lindenlaub et al., 2024). Assuming firms are atomistic ex-ante excludes the Robinson (1933) economy of company towns, and other geographically localized forms of monopsony power as discussed by Manning (2003).

<sup>9</sup>A firm does not need to employ all types of worker  $k$ . Firm production function outlined in Section 4.3

<sup>10</sup>There is evidence that straight-line distance is a good proxy for travel time. Using data from New York State Department of Transportation, Phibbs and Luft (1995) find a correlation of 0.98 between straight-line distance and travel times, the number drops to 0.82 for distances below 15 miles such that travel time per mile is generally larger than for long distances.

in its wage schedule would affect, not only the quantity of workers it attracts, but also from where, in relation to its location, does it attract those extra workers. If firms cast a wide geographical net in terms of hiring, the marginal worker would be overly sensitive to wage changes, increasing the average firm-specific labor supply elasticity and hence wages.

**Strategic Interactions & Local Labor Markets** I do not take a stance on the geographical borders of labor markets, but rather allow them to be porous to the extent of workers' commuting costs. That is, regardless of a given worker's residential location she can always choose from the full menu of employers, but all else equal, further work commutes have lower average utility. This differs from other papers in the literature that assume labor markets are geographically segmented. Labor market concentration indices derived from such models do not capture the true set of outside options when workers can travel beyond their defined segment.<sup>11</sup> As such, firms compete for workers from *every* residential location,  $z$ , and aggregate *firm-location* specific labor supply curves to obtain production relevant *firm specific* labor supply for each type  $k$ . To sum, due to commuting costs firms are horizontally differentiated, not only by idiosyncratic preferences, but also due to distance from worker specific residential location. This means changes in firms' wages differentially affect labor supply across neighborhoods.

Given the highly granular nature of geographic competition, firms are not assumed atomistic. In each neighborhood  $z$ , firms compete in an oligopsonistic fashion with other firms in their own sector,  $g$ , and other employers. This builds on [Chan et al. \(2024\)](#) with the main difference being that worker types can vary by their residential location,  $z$ , which is unobserved by the firm, and type  $k$ , which is observed. In other words, the model is a smoothed version of company towns in [Robinson \(1933\)](#), where in the limit a large degree of geographic isolation, or sufficiently high commuting costs, can lead to a single monopsonist. From a policy perspective, to see why assuming away a specific definition of a local labor market is appealing note the following two examples. One, if antitrust authorities screen mergers based on measures such as HHI, then getting the *right* set of outside options for workers is crucial, and worker mobility must be captured. Two, to assess the wage effects of improved transit we must counterfactually allow workers to travel to previously infeasible locations. Models with segmented labor markets can only speak to this topic across CZs, yet many infrastructure projects are within.

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<sup>11</sup>For example, in the US [Azar et al. \(2020\)](#) define a labor market as the intersection of a Commuting Zone (CZ) with 6-digit occupational code, while [Lamadon et al. \(2022\)](#) defines them as the intersection of CZs with 2-digit NAICs codes. There are a few exceptions that allow labor markets to be geographically porous ([Pérez et al., 2022](#); [Datta, 2023](#); [Schubert et al., 2024](#))

## 4.2 Workers' Problem

### 4.2.1 Preferences

In each period,  $t$ , worker  $i$  of skill  $k$  first draws a preference shock  $v_{it}$  over neighborhoods  $\mathcal{Z}$ . Then, after making their residential decision, they receive a preference shock over the set of all possible employers  $\mathcal{J}$  given by the vector  $\epsilon_{it}$ . They receive this shock over the set of all employers regardless of their residential location. This timing assumption is similar to Quantitative Spatial Models with the exception that I allow workers to choose residential location and employer, as opposed to only work location in an Armington model of production (Tsivanidis, 2022; Couture et al., 2024; Baum-Snow and Han, 2024). Workers indirect utility function is given by the following:

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$$U_{izjt} = \ln\left(\frac{a_{zt}}{r_{zt}^\gamma}\right) + \frac{1}{\varphi} v_{izt} + \underbrace{\ln w_{kjt} + \ln u_{zkt} - \eta_k d_{z(i),o(j)} + \frac{1}{\beta_k} \epsilon_{ijt}}_{\text{worker } i \text{ compensation net of commuting costs at firm } j} \quad (3)$$

After  $v_{it}$  is revealed workers make a location decisions given the group  $k$  specific amenity,  $a_{zkt}$ , from residing in neighborhood  $z$ , and the unit price of housing,  $r_{zt}$ . Where  $\gamma$  is the the share of expenditure on housing by households, and  $\varphi$  is a scale parameter. The stochastic residential preference shocks  $v_{izt}$  follow a nested logit distribution with a *homogenous* within nest correlation parameter  $\sigma$  given by the following distribution:

$$F(v_{izt}) = \exp \left\{ \sum_{c \in C} \left( \sum_{z \in \mathcal{Z}(c)} e^{-\frac{v_{izt}}{\sigma}} \right)^\sigma \right\} \quad (4)$$

Given residential location choices, idiosyncratic labor market preferences,  $\epsilon_{ijt}$ , are revealed and workers make employment decisions given the following. One, the type  $k$  specific vector of posted wages by firms,  $w_{kjt}$ . Two, the type-location specific vector of non-pecuniary benefits,  $u_{zkt}$ . This denotes the non-pay amenities for type  $k$  workers residing in neighborhoods  $z$  from working at firm  $j$ . Importantly, this is assumed orthogonal to commuting distance. Finally, given residential location,  $z$ , workers consider the commuting distance to each possible employer  $j$ , as denoted by  $d_{z(i),o(j)}$ . Workers can also decide to remain non-employed, in which case they receive  $w_{k0t}$ , and do not commute by definition. Average utility from nonemployment is normalized to zero. The parameter  $\beta_k$  measures the importance of wages relative to non-pay amenities. A higher value of  $\beta_k$

<sup>12</sup>This is derived from Cobb-Douglas preferences over housing and consumption, with housing expenditure share given by  $\gamma$



means employers are less horizontally differentiated, increasing workers' labor supply elasticity, and reducing firms' labor market power. The parameter  $\eta_k$  measures workers' marginal rate of substitution between commuting distance and wages. A higher value implies workers require higher compensation to be indifferent between their current job, and the next best alternative. In each period labor market stochastic shocks,  $\epsilon_{ijt}$ , denote worker's idiosyncratic preference to work at firm  $j$  and follows a nested logit with the following distribution function:

$$G(\epsilon_{ijt}) = \exp \left\{ \sum_{g \in \mathcal{G}} \left( \sum_{j \in J(g)} e^{-\frac{\epsilon_{ijt}}{\lambda_{kg}}} \right)^{\lambda_{kg}} \right\} \quad (5)$$

Each nest,  $\mathcal{J}(g)$ , is defined to contain all employers in an industry  $g \in \mathcal{G}$ . The type-industry specific parameters  $\lambda_{kg} := \sqrt{1 - \text{corr}(\epsilon_{ikj}, \epsilon_{ikl})}$  measure the degree of independence between preference shocks for firms within the same nest. A higher value implies more independence and less correlation. As  $\lambda_{kg} \rightarrow 1$  firms within a nest are at their maximal degree of horizontal differentiation, and are perceived as independent alternatives. Workers have relatively inelastic labor supply elasticities, and firms' market power is high. On the other hand, as  $\lambda_{kg} \rightarrow 0$  they are perceived as perfect substitutes, eliminating any within nest horizontal differentiation. This means that, within nest, any small change in wages is sufficient to cause workers to move to another employer, implying higher labor supply elasticities and lower firm market power. The parameters  $\beta_k, \lambda_{kg}, \eta_k$  govern firm specific labor supply elasticity with respect to wages as will be shown in section (4.2.2)

In this model, firm specific labor supply curves are upward sloping due to horizontal differentiation arising from two sources. One, the idiosyncratic preference term which captures the stochastic preference over amenities. Two, which is the contribution of this paper, the worker-firm specific distance term. As the firm expands along its labor supply curve, it attracts workers from different neighborhoods, who with higher wages, are now more willing to commute. Note that an important assumption is that employers can only observe the *average labor supply curve for a given group k*, and therefore cannot discriminate by worker residential neighborhood,  $z$ . Due to disutility from commuting, workers who commute further are more sensitive to wage changes, reducing the firms' ability to markdown wages. An empirical question is to what extent workers' sensitivity to commuting is sufficient to offset wage setting power (Mas, 2025).

#### 4.2.2 Labor Supply & Residential Sorting

In each period,  $t$ , workers observe the vector of neighborhood amenities  $\{a_{zkt}\}$ , housing rents  $\{r_{zt}\}$ , the set of firms  $\mathcal{J}$  and firms' compensation packages denoted by  $\{w_{kjt}, u_{kjt}, u_{zkj}\}$ . Given these values workers then draw their idiosyncratic preference shocks over neighborhoods,  $\nu_{izt}$ ,

and choose where to locate by forming expectations over idiosyncratic labor market preference shocks. Given their residential choice, labor market preferences over employers,  $\epsilon_{ijt}$ , are revealed, and workers choose at which firm to supply their labor. I solve the workers' problem backwards by deriving firm-specific labor supply, *given residential locations*, then I show how workers make these residential locations by forming expectations over labor market outcomes to derive residential choice probabilities. Worker  $i$  of type  $k$ , living in city  $c$  and neighborhood  $z$ , chooses amongst employers  $j \in \mathcal{J}$  to maximize  $\tilde{U}_{ijt}$  as denoted by (3)

$$\tilde{U}_{ijt} = \underbrace{\ln w_{kjt} + \ln u_{zkjt} - \eta_k d_{z(i),o(j)}}_{=: v_{zkjt}} + \frac{1}{\beta_k} \epsilon_{ij} \quad (6)$$

Workers who choose to remain in non-employment have a utility given by

$$U_{ik0t} = \ln w_{k0t} + \frac{1}{\beta_k} \epsilon_{ik0} \quad (7)$$

where  $w_{k0t}$  are unemployment benefits to be exogenously set. I assume that commuting distance when unemployed is zero. Note the assumption that non-employment average utility is location invariant.

**Neighborhood-Firm specific Labor Supply** Following from (6), and given the nested logit assumption on preferences, the share of workers from neighborhood  $z$  of type  $k$  working at firm  $j$  from nest  $\mathcal{J}(g)$  is given by:

$$s_{zkjt} := s_{zkg t} \underbrace{\frac{\exp\left(\frac{\beta_k v_{zkjt}}{\lambda_{kg}}\right)}{I_{zkg t}}}_{s_{zkj|g t}} \quad (8)$$

Where  $s_{zkg t}$  is the share in industry  $g$

$$s_{zkg t} = \frac{I_{zkg t}^{\lambda_{kg}}}{\sum_f I_{zkft}^{\lambda_{kf}}} \quad \text{and} \quad I_{zkft} := \sum_{l \in \mathcal{J}(f)} \exp\left(\frac{\beta_k v_{zkft}}{\lambda_{kg}}\right)$$

The labor supply elasticity of workers in neighborhood  $z$  of type  $k$  to firm  $j$  is given by: <sup>13</sup>

$$\mathcal{E}_{zkt} := \beta_k \left[ \frac{1}{\lambda_{kg}} + \left( 1 - \frac{1}{\lambda_{kg}} \right) s_{zklgt} - s_{zkt} \right] \quad (9)$$

Equation (9) denotes several mechanisms. One, as  $\beta_k \rightarrow \infty$  horizontal differentiation between firms vanishes and workers' labor supply to any particular firm becomes infinitely elastic, eroding firms' market power. Similarly as  $\lambda_{kg} \rightarrow 0$  firms are viewed as perfect substitutes within each nest, increasing the perceived number of different alternatives for workers, and increasing firm-specific labor supply elasticity. At the other extreme, as  $\lambda_{kg} \rightarrow 1$  we are in the logit case and elasticities can be written as  $\beta_k (1 - s_{zkt})$ . Two, firms with high labor market shares from neighborhood  $z$  face lower labor supply elasticities *from that location* since  $\left( 1 - \frac{1}{\lambda_{kg}} \right) s_{zklgt} - s_{zkt} \leq 0$ . Whether the neighborhood invariant elasticity labor supply is lower too depends on the geographical distribution of the firms' workforce. If all workers are from neighborhood  $z$ , then the change is one-to-one. Otherwise, it depends on whether the change in the neighborhood-firm specific elasticity (marginal) is lower than the within firm elasticity (average). To sum, firms affect workers' labor supply elasticities by offering better compensation packages, but this differentially impacts workers in different locations due to commuting costs, which means the neighborhood specific shares  $s_{zkt}$  do not change uniformly.

**Residential Sorting & Housing Demand** Since workers make residential choices before observing idiosyncratic labor market preferences, residential choice is shaped by residential amenities, housing costs, and *expected labor market compensation*. Given the distributional assumptions, the latter is given by

$$\begin{aligned} RMA_{zkt} &= \mathbb{E} \left[ \max_{j \in \mathcal{J}} \{v_{zkt} + \epsilon_{ij}\} \right] \\ &= \ln \left[ e^{\beta_k v_{zkt}} + \sum_g \left( \sum_{j \in N_g} e^{\frac{\beta_k v_{zkt}}{\lambda_{kg}}} \right)^{\lambda_{kg}} \right] \end{aligned} \quad (10)$$

The term  $RMA_{zkt}$  summarizes access to employment opportunities from residential neighborhood  $z$ , for workers of type  $k$  at time  $t$ . The last equality follows from the distributional assumption on idiosyncratic labor market preferences (McFadden et al., 1978). Therefore, after drawing location preferences workers choose neighborhood  $z$  to maximize the following

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<sup>13</sup>Labor supply elasticity conditional on location is derived in Appendix B.2

$$\max_{z \in \mathcal{Z}} \underbrace{\ln a_{zkt} - \gamma \ln r_{zt} + RMA_{zkt}}_{y_{zkt}} + \frac{1}{\varphi} v_{zit} \quad (11)$$

Given the distributional assumption in (4), the share of workers residing in neighborhood  $z$  is given by:

$$\mu_{zkt} := \pi_{kct} \underbrace{\frac{\exp\left(\frac{\varphi y_{zkt}}{\sigma}\right)}{\Phi_{ckt}}}_{\pi_{zk|ct}} \quad (12)$$

where

$$\pi_{zkct} = \frac{\Phi_{ckt}^\sigma}{\sum_{c'} \Phi_{c'kt}} \quad \text{and} \quad \Phi_{ckt} = \sum_{z' \in \mathcal{Z}(c)} \exp\left(\frac{\varphi y_{z'kt}}{\sigma}\right)$$

The number of workers residing in neighborhood  $z$  of type  $k$  at time  $t$  is therefore given by  $\ell_{zkt} := m_{kt} \mu_{zkt}$ . Note that since firms are not assumed to be atomistic, a change in a single firm's wage schedule can affect the population distribution of a given neighborhood. This allows for potential population declines following localized labor demand shocks, for example the demographics in factory or mining towns can drastically change after plant closures (Bound and Holzer, 2000; Notowidigdo, 2011; Krueger, 2017). This example shows that such effect depends on employer size in a given city or neighborhood. Formally, define  $\zeta_{zkjt}$  to be the population elasticity for type  $k$  workers in neighborhood  $z$  with respect to firm  $j$  wages at time  $t$ . Given the probabilities in (12) and the residential market access term in (10) this elasticity has the following form:<sup>14</sup>

$$\zeta_{zkjt} = \varphi \beta_k s_{zkjt} \left[ \frac{1}{\sigma} + \left(1 - \frac{1}{\sigma}\right) \pi_{zk|ct} - \pi_{zkct} \right] \quad (13)$$

Finally, given employer choice for all residents in neighborhood  $z$ , demand for housing has the following form:

$$H_{zt}^d(r_{zt}) = \frac{\gamma}{r_{zt}} \underbrace{\sum_{k \in \mathcal{K}} m_{kt} \mu_{zkt} \sum_{j \in \mathcal{J}} w_{kjt} s_{zkjt}}_{\text{Total Earnings in neighborhood } z := E_{zt}} \quad (14)$$

The cobb-douglas preference specification implies that aggregate housing demand in a given neighborhood  $z$  is a fraction of total earnings received by residents.

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<sup>14</sup>Derivation in Appendix B.2

### 4.3 Firms' Problem

The other side of the labor market is characterized by a set of heterogeneous firms  $\mathcal{J}$  and a wage-posting framework. The location of each firm  $j$  is assumed exogenous, and is given by  $o(j) \in \mathcal{Z}$ . In each period, firms observe the labor supply curve of each type  $k$  (*firm-specific* labor supply), but cannot observe the residential location of each worker. In other words, firms only observe the aggregate of equation (8) across neighborhoodss for each type  $k$ . Therefore for a given firm  $j$  the labor supply curve of workers with type  $k$  is given by  $\ell_{kjt} = \sum_{z \in \mathcal{Z}} \ell_{zkt} s_{zkjt}$ . Firms also observe their production function  $F_{jt}(\ell_{.j})$ , where  $\ell_{.j} = (\ell_{1jt} \dots \ell_{kjt})$  and an exogenous output quantity,  $Q_{jt}$ . Firms then decide on the number of workers  $\ell_{kjt}$  to hire by setting *type specific wages*,  $w_{kjt}$ , to minimize their wage bill.<sup>15</sup> Formally, in each period  $t$ , firm  $j$  sets wages  $\{w_{kjt}\}$  to minimize their wage bill, taking as given other firms' wages, and type-specific labor supply functions. Assuming that output quantity  $Q_{jt}$  is exogenously given, the firm solves the following:

$$\begin{aligned} \min_{w_{kj}} \quad & \sum_{k \in \mathcal{K}} w_{kjt} \ell_{kjt} \\ \text{s.t. } \quad & F^j(\ell_{.jt}) \geq Q_{jt} \\ & \ell_{kjt} = \sum_{z \in \mathcal{Z}} \ell_{zkt} s_{zkjt} \quad \forall (k, j) \in \mathcal{K} \times \mathcal{J} \\ & w_{kjt} \geq 0 \end{aligned} \tag{15}$$

For each firm  $j$  the production function follows the functional form below:

$$F_{jt}(\ell_{.j}) = \theta_{jt} \left( \sum_{k \in \mathcal{C}_j} \gamma_{kjt} \ell_{kjt}^\rho \right)^{\frac{\alpha_{jt}}{\rho}} \quad 0 < \alpha_{jt} \leq 1 \tag{16}$$

The parameter  $\theta_{jt}$  is a hicks-neutral firm specific TFP. The parameters  $\gamma_{kjt}$  are labor augmenting TFP shocks, and govern within firm relative productivity of workers. More importantly, they allow for sorting based on match-specific production complementarities between firms and worker types. The parameter  $\alpha_{jt}$  represents firm-specific returns to scale. Finally, *homogenous*  $\rho$  captures the degree of substitutability between workers of different types  $k$ . A value of  $\rho < 1$  indicates that workers are complimentary in production. Note that in the presence of diminishing returns, all else equal, bigger firms have lower marginal product of labor and therefore offer lower wages. A firm with better amenities, which is able to attract relatively more workers, will offer lower wages. This means that wages have a compensating differential component. Note that this is not the case under

<sup>15</sup>The assumption of exogenous quantities follows from assuming firm output being set of perfectly substitutable freely tradable goods

a constant returns to scale production function, where only worker productivity enters into wage determination.

Since the firm does not observe the residential location of each worker the relevant labor supply elasticity for the firm wage setting problem is the weighted average of elasticities across all neighborhoods, denoted by  $\mathcal{E}_{kjt}$ . Where the weights correspond to the share of workers from each neighborhood *within* the firm,  $\omega_{zkjt}$ . Thus the firm specific labor supply elasticity of type  $k$  to firm  $j$  is given by:<sup>16</sup>

$$\mathcal{E}_{kjt} = \sum_{z \in \mathcal{Z}} \omega_{zkjt} (\mathcal{E}_{zkjt} + \zeta_{zkjt}) \quad (17)$$

Equipped with the labor supply elasticity, the first order conditions of problem (15) yield the following Lerner condition for each firm  $j$ :

$$w_{kjt} = \underbrace{\lambda_{jt}}_{\text{shadow price of output}} \underbrace{\frac{\partial F^j(\ell_{\cdot j})}{\partial l_{kjt}}}_{MPL_{kjt}} \times \underbrace{\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt} + 1}}_{mdn_{kjt}} \quad (18)$$

Equation (18) yields the standard Lerner condition for wages as a markdown  $mdn_{kjt} \leq 1$  below the marginal product of labor, which is the product of the shadow price of output  $\lambda_{jt}$  and marginal product of labor. Note that unlike models of monopsonistic competition, labor supply elasticities,  $\mathcal{E}_{kjt}$ , are a function of labor supply parameters  $\beta_k, \lambda_{kg}$ , as well as labor market shares  $s_{kjt}$  (Berger et al., 2022; Chan et al., 2024). Further, horizontal differentiation due to commuting distance means elasticities are shaped by disutility from commuting,  $\eta_k$ , and the geographical distribution of workers relative to the firm. For example, if for a given firm workers live in close proximity then labor supply is relatively inelastic, as little compensation for commuting is needed. However, if a given worker lives further away they need to be compensated for their commute, and will be more sensitive to wage changes. Since the firm can only set wages given the average elasticity of its workforce this raises the wage set by the firm for all workers. This is akin to screening models where workers' outside options are unobserved to the firm, even after conditioning on observable types (Kline, 2025; Mas, 2025). The case of company towns as in Robinson (1933) is reached when residential locations and employers are sufficiently geographically isolated. In this case labor supply elasticities are low due to the limited number of employers within commuting distance, and firms can to a large degree act as a single monopsonist within a given geographic area.

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<sup>16</sup>Full derivation of below equation is given in Appendix B



## 4.4 Equilibrium

In each period  $t$ , given neighborhood and firm fundamentals  $\{\Theta_{z_i}, \Theta_{j_i}\}$  and the distribution of workers  $\{m_{kt}\}$ , where the former denote housing supply, residential amenities, firm specific non-pay amenities, and firm location, an equilibrium is defined by an allocation of workers to neighborhood-firm pairs  $\{m_{z,j|kt}\}$  and a set of wages  $\{w_{kjt}\}$  and rental prices  $\{r_{zt}\}$  such that in each period:

1. Workers choose neighborhood and employer pair  $(z, j)$  to maximize (3)
2. Given their locations, and workers' labor supply curves, wages  $\{w_{kjt}\}$  solve firms' problem given by (15)
3. Rental prices  $\{r_{zt}\}$  and wages  $\{w_{kjt}\}$  are such that housing demand, as given by (14), equals housing supply in each neighborhood
4. Consistency & population constraints:
  - (a) Firms' optimal labor inputs are consistent with the implied spatial distribution

$$s_{kjt}(\vec{w}; \vec{r}) = \sum_{z \in \mathcal{Z}} \overbrace{\frac{m_{z,j|kt}}{\sum_j m_{z,j|kt}}}^{s_{zjkt}} \underbrace{\sum_j m_{z,j|kt}}_{\mu_{zk}}$$

- (b) Population constraints:  $\sum_{j \in \mathcal{J}} s_{kjt} + s_{k0t} = m_{kt}$  and  $\sum_{z \in \mathcal{Z}} \mu_{zkt} = m_{kt} \quad \forall k \in \mathcal{K}$

Equilibrium existence follows the fixed-point argument in [Chan et al. \(2024\)](#) Appendix B. Uniqueness however is only guaranteed under a set of sufficient condition. Assuming the only component that horizontally differentiates firms are the unobservable idiosyncratic preferences, then under any log-concave distribution for worker's stochastic, for example Type I Extreme Value, we have a unique equilibrium whenever. This is proved in [Chan et al. \(2024\)](#) showing that a unique equilibrium exists whenever the the super-labor supply elasticity, and the cross super labor supply elasticities satisfy the gross substitutes property. Which holds under log-concavity. However, when there are two sources of horizontally differentiation, observable commuting costs, and stochastic preferences, the sum of these two need not be log concave. And additional restrictions need to be imposed to satisfy the assumptions in [Chan et al. \(2024\)](#). I provide a more detailed derivation of the semi super elasticity in Appendix B.5. Conceptually, uniqueness is guaranteed when firm labor supply is concave, that is workers are more inelastic as the firm grows. However, with commuting costs the firm grows by hiring more workers from a given neighborhood, but also workers living

farther away. The latter require more compensation due to their longer commute. As such, firm growth changes the composition of workers, and therefore the average labor supply elasticity.

#### 4.5 Residential Market Access & Concentration Index

A novel feature of nesting classical monopsony into a quantitative spatial model is a generalized concentration index at the neighborhood-type level. Building on [Chan et al. \(2024\)](#), and given the spatial distribution of workers, the generalized concentration index for neighborhood  $z$  and workers of type  $k$  is given by

$$GCI_{zkt} = \left[ \prod_{g \in \mathcal{G}} \left( \exp \left\{ \sum_{j \in \mathcal{J}(g)} s_{zkt|g} \ln s_{zkt|g} \right\} \right)^{s_{zkt|g} \lambda_{kg}} \right] \times \left[ \exp \left\{ \sum_{g \in \mathcal{G}} s_{zkg} \ln s_{zkg} \right\} \right] \quad (19)$$

If workers from a given neighborhood are all employed at one employer the concentration index equals to 1. Importantly, the concentration index is at the neighborhood level offering a model based measure of labor market power at granular spatial units.

Importantly, there is a tight connection between neighborhood level concentration  $GCI_{zkt}$  and neighborhood level residential market access  $RMA_{zkt}$ . More precisely, residential market access term can be decomposed into two components. One, the “matching component” which captures the systematic level of utility for workers in a given neighborhood from matching with a set of employers. Second, is the “concentration component” which captures the extent of labor market concentration at the neighborhood level. More precisely

$$RMA_{zkt} = \sum_{j \in \mathcal{J}} v_{zkt} s_{zkt} - \ln GCI_{zkt} \quad (20)$$

Note that changes to firm productivity, commuting costs, or any other fundamentals change both concentration and the matching component simultaneously. This happens due to changes in sorting and firm wage setting responses. All else equal labor market concentration acts as a population shifter and therefore has an inverse relationship with housing rents. The model therefore offers a microfoundation for the empirical relationship between housing costs, and labor market concentration documented by [Kahn and Tracy \(2024\)](#).

### 5 Identification & Estimation

In this section I outline the identification & estimation strategy used to recover labor supply and demand parameters. I start by describing the identification of labor supply preferences, this is

broken down into the identification of the distribution function  $G(\cdot)$  which governs workers idiosyncratic preferences, as summarized by  $\beta_k, \lambda_{kg}$ . Equipped with these parameters I then outline how to identify workers marginal disutility from commuting  $\eta_k$  using a modified gravity equation. Given the identification of labor supply parameters, the Lerner condition in equation (18) allows the recovery of marginal product of labor, which along with variation in the ratio of worker types is used to identify the substitution parameter  $\rho$ . With this in hand, other firm-specific labor demand parameters are identified. Residential demand parameters will be calibrated outside the model. I calibrate residential choice preferences using parameter estimates from [Baum-Snow and Han \(2024\)](#) and [Couture et al. \(2024\)](#). Specifically, I set  $\varphi = 3$  and  $\sigma = 0.35$ . This takes into account the mapping from the Frechet distribution in those papers, to a nested logit, with a homogenous nesting parameter  $\sigma$

## 5.1 Labor Supply Parameters

### 5.1.1 Idiosyncratic Scale & Correlation Terms: $\beta_k, \lambda_{kg}$

The identification strategy of the idiosyncratic labor supply preferences distribution  $G(\cdot)$  follows [Chan et al. \(2024\)](#) with the main difference being in the definition of worker types to include their residential locations. Following [Berry \(1994\)](#) the quasi-supply function of equation (8) is defined as the log ratio of the share of type  $k$  workers, residing in neighborhood  $z$ , who work at firm  $j$  at time  $t$ ,  $s_{zkt}$ , relative to their neighbors of the same type who are in non-employment,  $s_{zk0t}$ . The quasi-labor supply function is given by:<sup>17</sup>

$$\ln \frac{s_{zkt}}{s_{zk0t}} = \beta_k \ln \frac{w_{kjt}}{w_{k0t}} + (1 - \lambda_{kg}) \ln s_{zklgt} - \beta_k \eta_k d_{z,o(j)} + \ln u_{zkt} \quad (21)$$

The identification challenge in estimating equation (21) is the correlation of earnings, neighborhood specific inside shares, and potentially distance to workplace,  $d_{z,o(j)}$ , with the unobserved amenities term. More precisely, an increase in the unobserved non-wage amenities simultaneously changes reduces offered wage, and increases the labor supply for the firm, leading to an inconsistently lower estimate of  $\beta_k$ . To address this I identify the model parameters using an internal instrumental variable approach following [Lamadon et al. \(2022\)](#). This entails varying wages through productivity changes, while holding non-pay amenities *relatively* constant.

**Assumption 1.** Labor augmenting TFP  $\gamma_{kjt}$  can be decomposed into a type specific and firm-type specific components, where the latter follows an AR(1) process. That is:

$$\mathbf{1a)} \quad \gamma_{kjt} = \bar{x}_{kt} x_{kjt}$$

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<sup>17</sup>Derivations for quasi-labor supply function are in Appendix [B.1](#)

**1b)** The firm-type specific component,  $x_{kjt}$ , follows an AR(1) process with persistence parameter  $\delta$  of the following form:

$$\ln x_{kjt} = \delta \ln x_{kjt-1} + \bar{\xi}_k + \bar{\xi}_{kjt}$$

**Assumption 2.** The firm-type specific amenities  $\ln u_{zkjt}$  follows an MA(1) process

In words, Assumptions 1 and 2 ensure that *changes* in amenities are transitory relative to *changes* in labor-augmenting productivity. Note that these two assumptions do not preclude the *levels* of amenities and productivity to be correlated. The differences in their transitory nature means that after taking sufficient differences over time, recent changes in productivity related variables (e.g. revenue and labor supply) are correlated with changes in wages, as they are a function of labor productivity, but are uncorrelated with changes in amenities, as they are transitory (Lamadon et al., 2022; Chan et al., 2024). Further, taking long differences of (21) eliminates the time invariant distance to workplace  $d_{z,o(j)}$ . The parameter  $\eta_k$  will be estimated using a Gravity equation as will be outlined in the next section.

**Proposition 1.** Under assumptions 1 and 2 *sufficient* long differences satisfy the relevance and exclusion restrictions *whenever the underlying composition of workers' unobservable skill is independent of changes in firm-level productivity shocks over the same time period*.

1. relevance:  $Cov(\gamma_{kjt+2} - \gamma_{kjt-3}, x_{kjt} - x_{kjt-1}) \neq 0$
2. exclusion:  $Cov(\ln u_{zkjt+2} - \ln u_{zkjt-3}, x_{kjt} - x_{kjt-1}) = 0$

Proposition 1 implies that changes over a long enough period in any productivity related variables, such as wages and shares  $s_{zkljt}$ , are affected by short period changes in other productivity related variables. However, changes in amenities over the same period, are orthogonal to such productivity shocks. Since quasi-labor supply is at the neighborhood-firm level, an important assumption is that neighborhood unobservable skill composition does not change over the same period over which the long difference is applied, conditional on type  $k$ . As such, at each period  $t$  take the difference in the quasi-labor supply of each  $(z, k)$  type of workers between periods  $t + 2$  and  $t + 3$  to obtain the following:

$$\Delta_{2,3} \ln \frac{s_{zkjt}}{s_{zk0t}} = \beta_k \Delta_{2,3} \ln \frac{w_{kjt}}{w_{k0t}} + (1 - \lambda_{kg}) \Delta_{2,3} \ln s_{zkljt} + \Delta_{2,3} \ln u_{zkjt} \quad (22)$$

Where  $\Delta_{2,3} h_t = h_{t+2} - h_{t-3}$ . For instruments to be valid they must correlated with long differences in earnings, inside share, but not with long differences in amenities term. Applying Proposition 1 this holds for any productivity related variables. In practice, equation (22) will be estimated

using 2SLS where the instruments to be used are one period changes in log revenue, neighborhood specific log inside share, and *firm-level* sum of the shares for other types employed by the firm. These instruments are valid whenever the firm level productivity shocks have a persistent effect on wages and shares, but not correlated with *changes* in long term amenities.

### 5.1.2 Marginal Commuting Disutility: $\eta_k$

To identify workers' commuting disutility I leverage within firm variation in the supply of workers *across* neighborhoods relative to variation in commuting distance to the firm. The exponential decay of the number of workers relative to distance to firm identifies commuting disutility for each type  $k$  under the assumption that firms are *spatially connected*, that is there is no one-to-one mapping from neighborhoods to firms. This is akin to the mobility assumption in AKM models for worker and firm wage effects. This is analogous to gravity equations used in the Quantitative Spatial Models literature, where variation in labor supply to origin destinations is used to identify disutility from commuting (Ahlfeldt et al., 2015; Tsivanidis, 2022; Dingel and Tintelnot, 2020). The main differences, however, is that having already identified the parameters  $\beta_k, \lambda_{kg}$  from the previous section, I can therefore separately identify commuting disutility from other labor supply parameters without the need to use the variation in earnings across space as in Ahlfeldt et al. (2015). To have a gravity equation that governs the flow of workers from neighborhoods to firms I impose the following separability assumption on the non-pay amenities term  $\ln u_{zkt}$ .

**Assumption 3.** Conditional on distance to workplace, non-pay amenities  $\ln u_{zkt}$  take the following separable form:

$$\ln u_{zkt} = \ln u_{kt} + \ln u_{zg(j)t} + \ln e_{zkt} \quad (23)$$

where

$$\mathbb{E}[e_{zkt} | d_{z,o(j)}, \Gamma_{kjgt}, \Gamma_{zkg}, k] = 1$$

In addition to permitting the identification of commuting disutility using gravity, Assumption 3, along with the previously identified labor supply parameters allows recovering firm specific amenities  $\ln u_{kt}$  from firm fixed effects in the gravity equation. Further note that the model does not exactly fit the data. After estimating the model I will carry validation checks with respect to commuting flows and the wage distribution to show that the baseline simulation of the model fits the data well.

**Proposition 2.** Whenever Assumption 3 holds the share of workers at neighborhood  $z$  of type  $k$

employed by firm  $j$  at time  $t$ , takes the following form:<sup>18</sup>

$$s_{zkt} = s_{zkt|gt} s_{zkt} = \exp \left\{ \tilde{\eta}_{kg} d_{z,o(j)} + \Gamma_{kjt} + \Gamma_{zkt} \right\} \tilde{e}_{zkt} \quad (24)$$

where

$$\begin{aligned} \tilde{\eta}_{kg} &:= \frac{\beta_k \eta_k}{\lambda_{kg}} \\ \Gamma_{kjt} &:= \frac{\beta_k (\ln w_{kjt} + \ln u_{kjt})}{\lambda_{kg}} \\ \Gamma_{zkt} &:= \ln \left( \frac{u_{zkt}^{\lambda_{kg}}}{I_{zkt}} \right) \end{aligned}$$

That is, a regression of log commute probabilities between each neighborhood-firm pair on firm and neighborhood fixed effects, plus commuting distance recovers an estimate of the parameter bundle  $\frac{\beta_k \eta_k}{\lambda_{kg}}$ . This parameter corresponds to the semi-elasticity of commuting flows with respect to distance.<sup>19</sup> *The identifying assumption is that conditional on firm and neighborhood fixed effects the unobserved factors that affect commuting in the error term are independent of euclidean distances between locations.* Figure 9 illustrates the identification argument graphically; the elasticity of commuting flows with respect to distance is identified off the covariance between the number of and distance to workers within each firm. In other words, conditional on firm productivity the decay in number of workers with distance identifies workers' willingness to pay for commuting. The faster the decay of in residualized log workers as distance increases, the greater the commuting costs. Note that even if workers of different types are observed to have the same commuting distribution, information on employer productivity is needed to identify the shape of their indifference curves. This revealed preference approach differs from [Le Barbanchon et al. \(2021\)](#) who has stated preferences on reservation wages and maximum travel distance, and is closer to the empirical method proposed by [Card et al. \(2024\)](#) who show that workers with higher commuting constraints should be observed to match with higher paying firms to compensate for their higher disutility.

I estimate equation (24) using separate flow-weighted commuting gravity regressions with firm and neighborhood fixed effects in 2016 for each  $(k, g)$  tuple. An empirical challenge in estimating commuting gravity equations is the presence of zero bilateral commuting flows in the data. This is especially the case when the number of firms or neighborhoods is large, creating a sparse com-

<sup>18</sup>Derivation of the model implied gravity equation is given in Appendix B.4

<sup>19</sup>This is equivalent to the parameter  $\kappa$  in [Ahlfeldt et al. \(2025\)](#). I do not report estimated values of this parameter but they range between  $-0.06$  and  $-0.09$

muting matrix. Although we have a continuous measure of workers, and the logit distribution is unbounded from above, implying there should always be a positive measure of neighborhood-firm pairs, there are zero flows in the data. One way to rationalize this is to assume infinitely prohibitive commuting costs between neighborhood-firm pairs with zero flows. As shown by [Dingel and Tintelnot \(2020\)](#) and [Redding \(2024\)](#) this is equivalent to the *calibrated shares* approach in International Trade ([Eaton et al., 2012](#)), and has the disadvantage of ruling out zero flows becoming positive. For example, if no workers from the west of a commuting zone are observed to work for a firm in the east, even under a reasonable commuting distance, any counterfactual will continue to assume these workers are constrained to their commuting radius given by the data, potentially underestimating welfare implications of policies such as transit infrastructure improvement, commuting subsidies, or changes in work arrangements such as work-from-home. Instead, I assume the zero flows in the data are due to having a finite sample from the continuous model, that is the population (sample) in each neighborhood is low relative to the number of potential employers. In which case the parameters of the model can be estimated using the Pseudo-Poisson Maximum Likelihood (PPML) estimator ([Silva and Tenreyro, 2006](#); [Correia et al., 2020](#)). Although, fixed-effect estimators in non-linear models potentially suffer from an Incidental Parameter Problem (IPP), the PPML estimator with two-fixed effects is asymptotically unbiased ([Fernández-Val and Weidner, 2016](#)). After recovering the fixed effects from regressions in equation (24) I use the labor supply parameters to recover  $\ln u_{kjt}$ . I outline the exact execution of this step in the estimation procedure.

## 5.2 Labor Demand Parameters

In this section I demonstrate how to use the estimated labor supply elasticities to identify firm production functions. Conditional on having firm level balance sheet data, where firm revenue is observed, the method to identify firms' production function can be split into two subparts. One, conditional on having recovered the elasticity of substitution parameter,  $\rho$ , other firm-specific parameters can be identified via inverting the firms' FOC in Equation (18). Two, absent endogeneity concerns, the substitution parameter is identified from variation in the ratio of worker types relative to changes in their marginal product. Under perfectly competitive labor markets the latter coincides with the wage ratio. However, due to wage markdowns this no longer holds, and marginal product of labor needs to first be identified.

### 5.2.1 Elasticity of Substitution: $\rho$

From Equation (18) we know that the wage is a markdown times the marginal revenue product of labor. Since labor supply parameters have been identified, firm-specific labor supply elasticity can be constructed using (4) and used to define the marginal revenue product of  $k$ -type workers at firm



$j$  as:

$$\tilde{w}_{kjt} := \frac{\mathcal{E}_{kjt} + 1}{\mathcal{E}_{kjt}} w_{kjt} \quad (25)$$

Fixing workers of type  $k, h \in \mathcal{K}$  the firm first order condition in (18), along with the production function functional form assumption 1 imply that

$$\ln \frac{\tilde{w}_{kjt}}{\tilde{w}_{hjt}} = (\rho - 1) \ln \frac{\ell_{kjt}}{\ell_{hjt}} + \ln \frac{\tilde{\gamma}_{kjt}}{\tilde{\gamma}_{hjt}} \quad (26)$$

Running OLS on (26) will provide biased estimates, since labor ratios are correlated with labor augmenting TFP. Using identification assumption 2 substitute in the innovation terms we get the following estimation equation:

$$\ln \frac{\tilde{w}_{kjt}}{\tilde{w}_{hjt}} = \zeta_{kht} + (\rho - 1) \ln \frac{\ell_{kjt}}{\ell_{hjt}} + \delta(\rho - 1) \ln \frac{\ell_{kjt-1}}{\ell_{hjt-1}} + \delta \ln \frac{\tilde{w}_{kjt-1}}{\tilde{w}_{hjt-1}} + \zeta_{khjt} \quad (27)$$

I estimate (27) using OLS after controlling for lagged labor marginal revenue product ratios for all possible types  $(k, h)$  within a given firm. That is, after correcting for markdowns using the estimated labor supply parameters, the inverse of the elasticity of substitution is identified off the covariance of wage and factor input ratios.

### 5.2.2 Other Parameters

Equipped with the substitution parameter I impose the following normalization assumptions to identify  $k$ -type specific labor augmenting TFP  $\gamma_{kjt}$  as well as the hicks-neutral TFP term  $\theta_{jt}$

**Assumption 4.** Normalize the labor-augmenting productivity terms for a given firm  $j$  and assume firms have no market power on the product market.

**4a)** Normalize the labor augmenting TFP at each firm such that:

$$\sum_{k \in \mathcal{C}_{jt}} \gamma_{kjt} = 1$$

**4b)** Output market is perfectly competitive  $\Rightarrow \lambda_{jt} = P_{jt}$

**Proposition 3.** Under Assumption 4 the following holds (Chan et al., 2024)

1. Given an estimate of  $\rho$  firm-specific labor augmenting TFP is identified from the cross-section by:

$$\gamma_{kjt} = \frac{1}{\sum_{h \in C_{jt}} A_{khjt}}$$

Where

$$A_{khjt} = \frac{\tilde{w}_{kjt}^{-1} \ell_{kjt}^{\rho-1}}{\tilde{w}_{hjt}^{-1} \ell_{hjt}^{\rho-1}}$$

2. To back out the returns to scale parameter invert the firms' FOC (18) to obtain:

$$\alpha_{jt} = \frac{\rho}{R_{jt}} \times \sum_{k \in C_{jt}} \frac{\tilde{w}_{kjt}}{\rho \ell_{kjt}^{\rho-1}} \times \sum_{k \in C_{jt}} \gamma_{kjt} \ell_{kjt}^{\rho}$$

3. Finally Hicks-neutral TFP identified up to scale

$$P_{jt} \theta_{jt} = \frac{R_{jt}}{\left[ \sum_{k \in C_{jt}} \gamma_{kjt} \ell_{kjt}^{\rho} \right]^{\frac{\alpha_{jt}}{\rho}}}$$

### 5.3 Estimation Procedure

1. For each  $k$ : Estimate labor supply parameters by running 2SLS on equation (22). Use as instruments one-period changes in log revenue *at the enterprise level*, from firms' balance sheet data in the NALMF, and one-period changes in log neighborhood-type-firm shares  $s_{zkjt}$ .  
20
2. For each  $(k, g)$  tuple: Estimate the gravity equation (2) using PPML with firm and neighborhood by industry fixed effects. Given the estimated  $\tilde{\eta}_{kg}$  and the estimated  $\beta_k, \lambda_{kg}$  recover  $\eta_k$ .  
21
3. From equation (2), given values of  $\beta_k, \lambda_{kg}$ , and firm fixed effects, the firm specific amenities  $\ln u_{kjt}$  could be recovered from the following equation

$$\ln u_{kjt} = \lambda_{kg} \underbrace{\Gamma_{kjgt}}_{\text{Firm FEs}} - \beta_k \ln w_{kjt}$$

<sup>20</sup>I run (22) with dummy variables for each industry to obtain industry specific  $\lambda_{kg}$

<sup>21</sup>In practice I recover multiple  $\eta_k$  since PPML is run at the industry level. I take the average value after multiplying and dividing by  $\eta_{kg}$  and  $\beta_k$  respectively.

4. Given estimated parameters and firm fixed effects, recover  $\ln u_{zkg t}$  from neighborhood specific employment log odds using the following equation

$$\ln \left( \frac{s_{zkg t}}{s_{zk0 t}} \right) = \ln u_{zkg t} + \lambda_{kg} \ln \left( \sum_{j \in} \exp \left\{ \frac{\beta_k \ln w_{kjt} + \ln u_{kjt}}{\lambda_{kg}} \right\} \right)$$

5. I use establishment specific shares  $s_{zkg t}$ , residential distribution,  $\pi_{zkg t}$ , the estimated labor supply preference parameters  $\{\beta_k, \lambda_{kg}\}$ , and the calibrated residential choice parameters  $\{\varphi, \sigma\}$  to construct  $\mathcal{E}_{zkg t}$ ,  $\zeta_{zkg t}$  and the appropriate weights to construct  $\mathcal{E}_{kjt}$  as presented in equation (17).
6. Given the constructed  $\mathcal{E}_{kjt}$ , I recover marginal revenue product of labor  $\tilde{w}_{kjt}$ , and estimate the elasticity of substitution parameter  $\rho$ , and persistence parameter  $\delta$  as outlined by estimation equation (27). Given the estimated  $\rho$  apply proposition (3) to recover  $\{\gamma_{kjt}, \alpha_{jt}, \tilde{\theta}_{jt}\}$
7. Construct  $RMA_{zkt}$  as outlined in Equation (10) for each neighborhood-type pair  $(z, k)$ . Given these values, use residential choice probabilities  $\pi_{zkg t}$ , and neighborhood specific housing rental indices to back out neighborhoods-type specific amenities  $\ln a_{zkt}$  using choice probabilities. The amenities are normalized with respect to neighborhood with the lowest population across all 50 CMAs

## 5.4 Estimation Results

### 5.4.1 Labor Supply & Demand Primitives

Estimates of group  $k$  specific labor supply parameters are presented in Table 3. As discussed in section 4.2.1, a lower  $\beta_k$  and higher  $\lambda_{kg}$  contribute to higher horizontal differentiation across employers leading to more monopsony power and therefore lower markdowns. Table 3 shows that younger workers have higher marginal utility of earnings, and therefore view employers as relatively more substitutable than older workers. The within nest correlation parameter  $\lambda_{kg}$  is estimated to be around 0.3, and relatively stable across groups. On the other hand, estimate of  $\eta_k$  shows significant heterogeneity across age and gender. Estimated values indicate that older male workers require an extra 2.2% increase in annual earnings to commute a further 1 km to work. Older female workers require a 2.6% increase in annual earnings, exhibiting a gender gap in commuting disutility of almost 20%. This is not true however for younger workers, where independent of gender, compensation required for an additional one kilometer of commuting is around 0.9% of annual earnings. The estimated gender gap is close to that reported by [Le Barbanchon et al. \(2021\)](#), however workers in my sample exhibit much higher commuting disutility. This is expected given

that these are full-time worker, whereas the sample in [Le Barbanchon et al. \(2021\)](#) consisted of unemployed workers only. Further, the estimated values of  $\eta_k$  are close to the values estimated by [Ahlfeldt et al. \(2025\)](#) who estimate origin-destination level gravity equations using commuting flows and travel time on a sample of full-time workers in Denmark. My estimates are close to theirs, and exhibit the same life-cycle pattern. Assuming average annual earnings of 84,000 for this group, this amounts to approximately \$1848 of compensation per year for an increase in distance to work by one kilometer. Assuming a work-year of 250 days this corresponds to an increase of \$7.40 per day. Further, assuming a one kilometer increase in travel distance corresponds to an increase in travel time by 3.4 minutes each way for a total of 6.8 minutes, older male workers require an increase in compensation corresponding to 1.55 times their hourly wage. For older female workers, with an elasticity of 2.7% their compensating differential for an extra 1 km of travel is 1.83 times the hourly wage. Together these results indicate that younger workers are more wage sensitive, and are less sensitive to commuting costs. On the other hand, older workers view employers as more differentiated, are less sensitive to wage changes, and put more weight on commuting distance when choosing between jobs. Further, amongst older workers there is a 20% gender gap in commuting disutility. This demonstrates that a primary advantage of including explicit commuting costs into worker preferences is identifying the role of commuting to work in explaining differences in job choice relative to other job attributes.

Table 5 shows the estimated labor demand parameters. Panel A shows the distribution of the firm-specific production function parameters  $(\theta_{jt}, \gamma_{kjt}, \alpha_{jt})$ . The average value of the returns to scale parameter across all firms is around 0.29, exhibiting high decreasing returns and significant heterogeneity across firms; with a 90-10 ratio of 8.14. The labor augmenting labor productivity  $\gamma_{kjt}$  is a within firm relative productivity measure and should be compared across groups within a given firm. Finally, the average  $\ln(\tilde{\theta}_{jt})$  is approximately 14.9 and varies significantly across firms. These values are close to firm production function estimation by [Tino \(2024\)](#) who applies a similar production functions estimation strategy for a set of Canadian firms using the CEEDD. Finally, estimates of elasticity of substitution in Panel B of Table 5 at 0.91 exhibits high levels of substitution across worker types. This result is consistent with estimates from [Chan et al. \(2024\)](#) and [Volpe \(2024\)](#), who although allow for heterogeneity across worker types and firms respectively, find an average EoS close to 0.9 with little variation across types and/or firms. Finally, estimates of  $\delta$  at 0.73 validates the high level of persistence in productivity shocks in Assumption 1.

#### 5.4.2 Elasticities & Markdowns

Table 4 provides employment-weighted average of firm labor supply elasticity for all workers, and by group  $k$ . Panel A shows that the average employment-weighted labor supply elasticity is around 7.5. This value is slightly higher than estimated values in the literature within the range

of 3-7. To better understand why this is the case, note that workers living further away from the firm are more elastic. This can be seen from the neighborhood-firm specific labor supply elasticity in equation (9) where the shares are decreasing with distance to employers. Therefore if the marginal worker lives farther away from the firm labor supply elasticity at the  $k$ -level will be higher. This reflects a key mechanism of the model; proximity to workforce affects the slope and intercept of firms' labor supply curves. To quantify the role of commuting in shaping firm labor supply elasticities I simulate the model under a counterfactual of no commuting costs, to show that they commuting explains 5% of the reported employment-weighted elasticity. The corresponding aggregate markdowns are 84.7% indicating that workers receive on average 85% of their marginal revenue product of labor. These averages hide significant heterogeneity across worker groups and firms. Panels B-D show that younger workers are more elastic and earn on average 91% of their marginal revenue product of labor, compared to only 79.3% for older workers. Note that although there is a gender gap in commuting costs for older workers, this does not necessarily show up in the average markdowns. To see why note that there are two forces at play. One, more productive firms offer lower markdowns, since independent of a worker's neighborhood, they have higher labor market shares at any level of geographical aggregation. Two, small firms with low aggregate shares are more likely to attract workers from nearby neighborhoods, and therefore also offer low markdowns. That is, what shapes a firms' markdown is the *average shares* from its' workforce neighborhoods. To better see this, note that under fixed spatial distribution of workers the firm specific labor supply elasticity could be re-written as:

$$\mathcal{E}_{kj} = \beta_k \left[ \frac{1}{\lambda_{kg}} + \left( 1 - \frac{1}{\lambda_{kg}} \right) \tilde{s}_{zkg} - \tilde{s}_{zkg} \right] =: \mathbb{E}[\mathcal{E}_{zkg}|j, k]$$

where

$$\tilde{x} = \sum_z x \mathbb{P}(z|j, k)$$

This equation nests the case where each firm is restricted to hire from a given CMA. Demonstrating that explicitly modelling spatial linkages between locations renders the proposed framework amenable at any aggregation level. Another implication of this equation is that within firm variation in where workers live affects firms' labor supply elasticity, highlighting that workforce composition can shape wage markdowns. This result resembles the findings in [Volpe \(2024\)](#) who shows that unobserved variation across workers in their relative valuation of wage and non-wage amenities allows even low productivity firms to have low wage markdowns, as they attract less wage sensitive workers. The framework proposed here assumes workers differ in their marginal rate of substitution between wage and non-wage amenities due to idiosyncratic observable differences in commuting costs.

### 5.4.3 Commuting & Firm Productivity

If workers tradeoff wages against commuting time in making job choices, it should be the case that in travelling farther away workers match with more productive firms. Table 6 presents results from regressing log firm TFP,  $\ln(\tilde{\theta}_{jt})$  on workers' commuting distance. Note that the recovered  $\ln(\tilde{\theta}_{jt})$  are independent of the estimated  $\eta_k$ , since only  $\beta_k$  and  $\lambda_{kg}$  are needed to recover the firm specific productivity parameters. Permitting this exercise to serve as a qualitative validation test for the estimated willingness to pay parameters. Estimates in column (1) show that an increase in commuting distance by 1 km is associated with an increase in firm TFP by 1%. With an average commuting distance of 11 km this corresponds to an elasticity of 0.1. Within CMA estimates in column (3) are similar but slightly lower. Columns (2) and (4) of Table 6 show that the firm TFP commuting premium is higher for women, validating the estimated willingness to pay for shorter commute estimates  $\eta_k$ . This is because workers with higher disutility from commuting, in this case women, require higher compensation to travel farther. An explanation for this is that female workers are more spatially mismatched. That is, they live farther away from high paying jobs. This is possible if married couples bargain over residential location, coupled with lower female bargaining power. These results are in line with Card et al. (2024) who use AKM fixed effects to show that longer commutes are associated with higher firm premiums. Note that with variable firm markdowns firm premia captures both firm specific markdowns and firm productivity.

## 6 Counterfactuals

I simulate the model using an underrelaxed Jacobi iteration algorithm to solve for equilibrium wages, rental prices and allocation to firms and neighborhoods using recovered housing supply, firm production parameters and given population for 2016 only. Refer to Appendix C for the detailed simulation algorithm. Note that the model does not fit the data exactly by construction. This is due to the separability assumption in equation (23). The simulated model matches the data well recovering a variance of log wages of 0.21 which is close to that of the data of 0.22. The simulated commuting distribution also matches the 2016 commuting distribution closely.

### 6.1 SkyTrain Simulation

To better understand the underlying mechanisms behind the SkyTrain expansion effect on earnings with respect to changes in worker-firm sorting, labor supply elasticities, and markdowns, I simulate the model for the *entire* economy in 2016 under a range of changes to commuting costs to emulate the absence of the Canada Line. Note that commuting costs in the model between neighborhood  $z$  and firm  $j$  enter parametrically as  $\kappa_{z,j} := e^{\frac{-\eta_k \beta_k d_{z,o(j)}}{\lambda_{kg}}}$ . Specifically, I simulate the model under the following commuting costs parameterization

$$\kappa_{z,j} = \begin{cases} Ae^{\frac{-\eta_k \beta_k d_{z,j}}{\lambda_{kg}}} & \text{if } z \in \text{CanadaLine and } o(j) \in \text{SkyTrain} \setminus \text{CanadaLine} \\ e^{\frac{-\eta_k \beta_k d_{z,j}}{\lambda_{kg}}} & \text{otherwise} \end{cases}$$

That is, I reduce commuting costs by a factor  $A$  between neighborhoods that gained access to the SkyTrain network and any other neighborhood that was on the SkyTrain prior to 2009. This corresponds to the dark blue and gray areas in Figure 2 respectively. Note that areas that did not gain access to the subway network could still be affected due to migration, employment responses, or reallocation between workers and firms across space. These correspond to the areas in yellow. I run simulations for multiple values of  $A$  to simulate the economy in absence of the Canada Line and compare it to the simulated model in 2016. Importantly, I simulate the model for the entire economy, not the Vancouver area only. This allows for migration across CMAs as a result of the Vancouver area offering a better balance between housing costs, firm access and commuting costs.

I report the percentage change in outcomes between the baseline economy, with the Canada Line, and the counterfactual economy, that takes as given the spatial distribution of firms in 2016, and assumes the Canada Line is removed. Outcomes are reported separately for three regions. First, neighborhoods that never had access to the SkyTrain network, corresponding to the yellow regions in Figure 2; Second, regions that have access to the Millenium or Expo lines, corresponding to the gray area in Figure 2; Finally, neighborhoods that gained access to the SkyTrain network due to the Canada Line, corresponding to the dark blue areas, and mainly consists of the suburb of Richmond.

Table 7 reports the percentage change in labor market outcomes, population and market concentration indices for  $A = 0.7$  corresponding to a 30% reduction in commuting costs between neighborhoods on the Canada Line and any other neighborhood where the Millenium and Expo Lines are located. First, average wages for workers on the Canada Line are 1% higher in presence of the Canada Line. At the same time, workers in other neighborhoods experienced a drop in earnings by 0.2 % for workers not on the SkyTrain network, and 0.3% for workers on the Expo & Millenium lines. Across all neighborhood types however, there is little change in markdowns. Instead wage effects seem to be driven by the resorting of workers on the Canada Line to more productive firms. To see that, note that under the Canada Line workers in affected neighborhoods experienced an increase of 5% in average employer TFP, as captured by  $\tilde{\theta}_{jt}$ . However, firm productivity is not captured by  $\tilde{\theta}_{jt}$  only, as firms differ in their returns to scale as well as their labor augmenting productivity. To assess how worker-firm sorting changes in presence of the Canada Line given the rich firm heterogeneity, I follow Chan et al. (2024) to assess how the composite term given by  $\tilde{\theta}_{jt} \alpha_{jt} \sum_{k \in \mathcal{K}} (\gamma_{kjt} \ell_{kjt}^\rho)^{\frac{\alpha_{jt}}{\rho} - 1}$  changes for each neighborhood type. Table 7 shows that workers on the Canada Line experienced an increase in this composite term by 1.7 %, indicating their



resorted to more productive firms. On the other hand, workers at other neighborhoods resorted to less productive firms with an average drop in the composite term by 0.34 %. This indicates that workers on the Canada Line resorted to more productive firms while crowding out workers from other areas. This shows that the transit projects could have negative spillovers on workers in far away neighborhoods due to increased firm access to other areas. This result is consistent with the findings of [Manning and Petrongolo \(2017\)](#) who show that improved transportation between an area of high and low unemployment can affect locals in high employment areas due to the stronger competition for jobs from new applicants. I further extend their findings by showing that competition between workers across neighborhoods occurs also on the intensive margin of firm productivity, as well as employment. One way to rationalize the change in worker firm sorting with little change in markdowns is to note that markdowns are at the firm-type level, not the neighborhood level. It is therefore possible that neighborhoods affected by a transportation shock reallocate to more productive firms who have lower markdowns for *all* workers.

Further, neighborhoods on the Canada Line experienced a population increase of 3.75% , whereas other neighborhoods saw a drop in their population. Note that the baseline population on the Canada Line is lower than other areas, indicating that workers changed their place of residence to benefit from the increased accessibility. Moreover, Canada Line neighborhoods experienced an increase in employment by 8%, while other neighborhoods saw a slight drop in employment. This is line with workers on the Canada Line crowding out workers from other areas, not only on the intensive margin of firm types, but also on the extensive margin of employment. Consistent with the increase in population, employment, and wages, and given the fixed housing supply assumption, average rents increased on the Canada Line neighborhoods by 8.5%, whereas rents dropped by 0.7% and 0.5% at neighborhoods not on the SkyTrain and Old station areas respectively.

Turning to welfare and labor market concentration, a key insight from this simulation is the fall in concentration for neighborhoods on the Canada Line by 8.6%. There are two drivers of this result. First is the drop in the share of non-employed for workers in these neighborhoods. As presented in Equation 19, all else constant, if the share of workers in some group  $g \in \mathcal{G}$ , including nonemployment drops, concentration index falls. This is assuming the newly employed workers have the same probabilities of sorting to firms. The second driver is the reallocation of workers across firms, presented by the increase in the employment weighted average of productivity of employers. The drop in labor market concentration in neighborhoods on the Canada Line was accompanied by a modest increase in concentration in other neighborhoods. Areas not on the SkyTrain network, and neighborhoods on Expo & Millenium lines experienced an increase in concentration index by 0.5% and 0.08% respectively. Although the spillovers in concentration across neighborhoods are modest, the results show that reducing labor market concentration in a set of neighborhoods, can be at the detriment of crowding out and increasing concentration in

other areas. It is important to note, however, that there is no directional relationship between the generalized concentration index and wages, instead both are determined in equilibrium by the allocation of workers to firms. Equation 20 indicates an inverse relationship between residential market access  $RMA_{zkt}$  and concentration  $\ln GCI_{zkt}$  at the neighborhood level. Table 7 exhibits this relationship by indicating an increase in market access for neighborhoods on the Canada Line by 3%. While neighborhoods not on the SkyTrain network, and those on the Expo & Millenium lines experienced a drop in market access by 0.2 % and 0.13% respectively. The passthrough from concentration index to market access is not one to one due to changes in equilibrium shares  $s_{zkt}$  and systematic utility  $v_{zkt}$ , due to changes in wages. The increase in population, employment, wages, and the fall in concentration index all contribute to the increased housing rents experienced by neighborhoods on the Canada Line. It is not clear, however, which factor is the primary contributor.

Overall, the results show that the Canada Line benefited workers in neighborhoods who gained access to the subway network. This was in tandem with negative labor market consequences for workers from other neighborhoods who were crowded out from high productivity employers, and culminating into tougher competition for jobs on the intensive and extensive margins.

## 6.2 Heterogeneous Firm Access & Spatial Variation In Markdowns

I use the model to examine the role of the geography of firms in shaping wage markdowns across space. To do so, I simulate the model under the counterfactual scenario of uniform distance between workers and firms, regardless of residential location. More presicely, I set commuting distance to 20 km for any worker firm pair that is within a 75 km radius of a given neighborhood. This is not necessarily a welfare improving shock since workers lose the option to work close to home, however within a given area it removes differential access to firms across neighborhoods.

Table 8 presens the change in variance of mean wages, markdowns, and marginal product across space for the four different group of workers. Results show that within a given CMA, the geography of firms explains a sizable share of the spatial variation in mean wages across neighborhoods. Focusin on male workers between the age of 40 to 60, the firm geography of firms explains 16.8% of variation in wages across neighborhoods. This rises to 42% within CMAs. The role of firm geography is more important in explaining earning variation across space for female workers in the same age group. This is due to their higher disutility from commuting, as such firm geography plays a more important role in shaping their earnings distribution. The effect on wages is primarily mediated through a reduction in the variance of marginal product of labor. Indicating that worker-firm geography plays an important role in shaping labor supply. Heterogeneous commuting costs explain around 40% of variation in earnings across neighborhoods within a CMA. Although the firm productivity are substantial in explaining the wage effect, the variance in markdowns across space also shrinks when commuting costs are equalized. The variance in markdowns

drops by 16 % on average across neighborhoods. The effects within and across CMA are not significantly different. Both the effects on markdowns and marginal product are mediated through changes in allocation of workers and firms. These results show that heterogeneous commuting costs explain a large share of the variance in markdowns across space, highlighting the role of equalizing firm access to eliminate inequality in labor market power across space.

## 7 Conclusion

Commuting costs are a classical source of market power. Alleviating workers' commuting burden increases the effective number of potential employers, and therefore improves workers' outside options. I leverage a subway expansion in Vancouver to show that workers who benefited from lower commuting costs experienced an increase in earnings relative to those who did not gain access. Empirical evidence, although suggestive, cannot rule out changes in workers' and firms' productivity due to the shock, and is silent on the general equilibrium effects of the shock across space. To decompose the mechanisms underlying the observed wage gains, and to investigate the general equilibrium effects of the subway expansion, I build, identify, and estimate a two-sided labor market matching model featuring strategic interactions in wage-setting model, commuting costs, and residential choice. The model extends the framework in [Chan et al. \(2024\)](#) to assess the role of commuting in shaping equilibrium outcomes, and extends the framework in [Caldwell and Danieli \(2024\)](#) to investigate worker to firm sorting in the presence of wage-setting behavior.

Simulation results show that workers in areas close to the subway expansion experienced an increase in earnings due to sorting to higher productivity firms, and faced lower labor market concentration. This was at the cost of workers from other areas, who faced higher competition for high-paying jobs, and slight increases in labor market concentration. To provide more evidence on the role of differential access to jobs in shaping the sorting of workers to firms, and shaping wage markdowns, I show that heterogeneous commuting costs can explain a large share of the spatial variation in wage markdowns, and marginal product of labor. Underscoring the role of horizontal commuting costs in shaping labor market power via sorting in equilibrium.

The proposed framework can be used to study the labor market consequences of any change in commuting costs, regardless of its source. For example, it could be used to better understand how work-from-home work arrangements affects the equilibrium allocation of workers to firms, and the resulting wage setting power ([De Fraja et al., 2022](#)). More generally, the proposed model can be used to study the effect of any spatially heterogeneous shock on labor market power while allowing for flexible commuting patterns ([Monte et al., 2018](#)).

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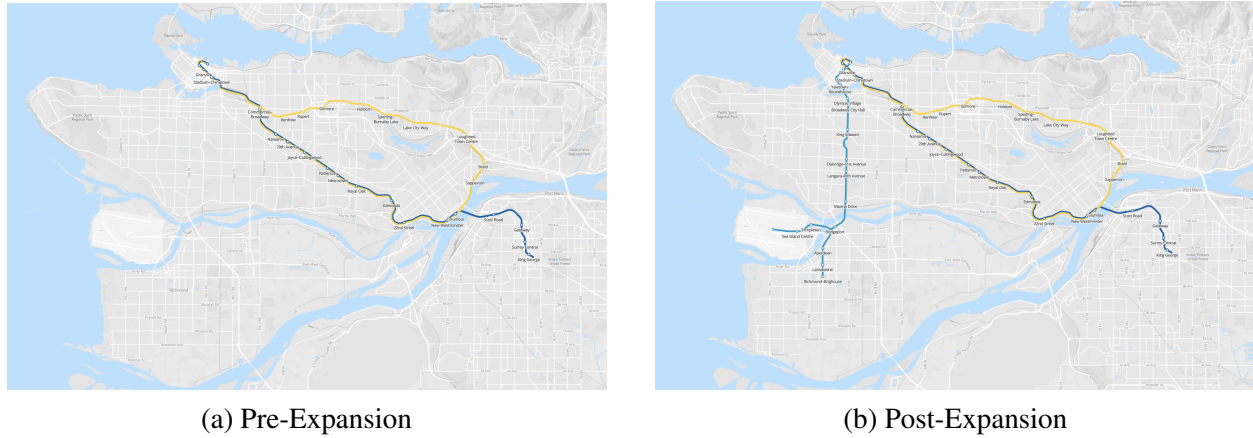
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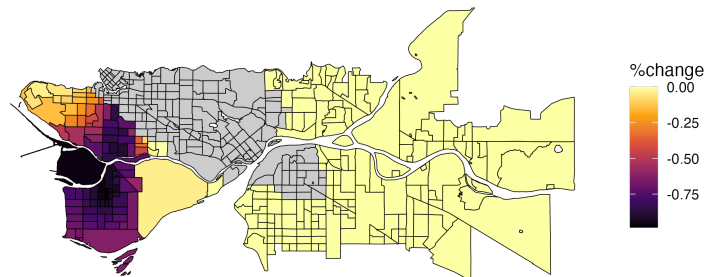
# Figures

Figure 1: Vancouver SkyTrain Maps Pre and Post Canada Line expansion



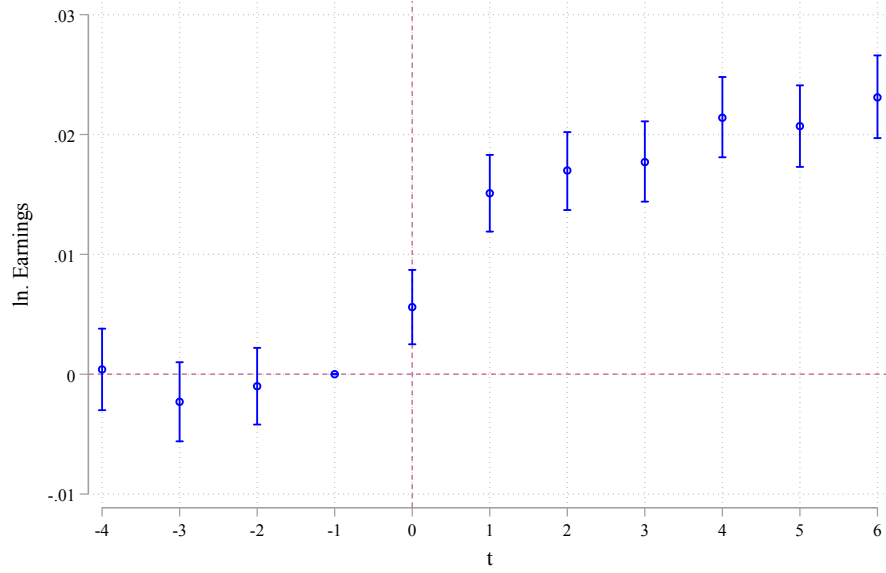
*Notes:* Panel A demonstrates the Vancouver SkyTrain in 2008 prior to the Canada Line expansion. At this point the SkyTrain system includes the Expo and Millennium lines. Built in 1986, and 2002 respectively. Panel B illustrates the Canada line expansion in light blue. The new line connects the airport in the west and Richmond in the south, with downtown Vancouver across the Fraser River.

Figure 2: Richmond & Rest of Metro Vancouver Public Transit Shares



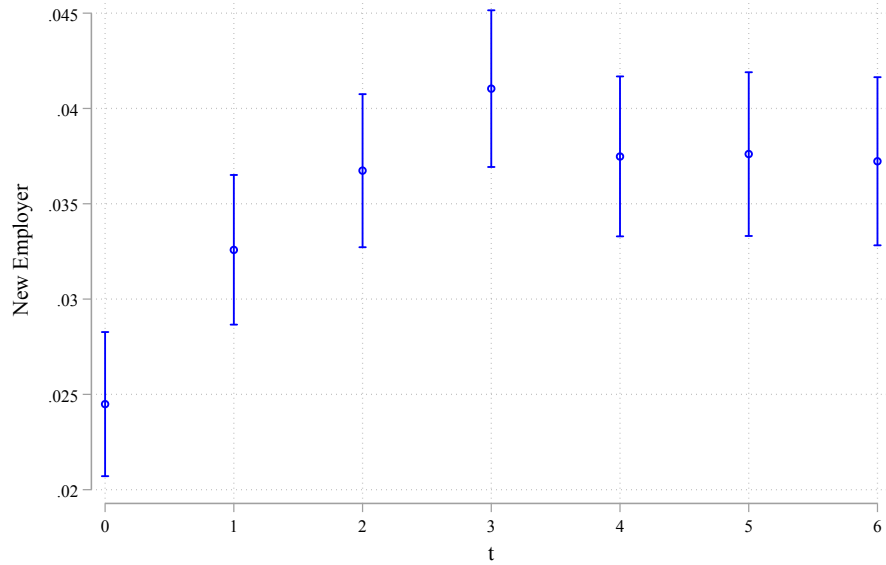
*Notes:* This Map illustrates treatment and control areas for the Vancouver SkyTrain expansion estimation. Darker blocks reflect a higher change in distance to nearest station during the expansion. Lighter colors reflect lower change. Neighborhoods in yellow represent the control group for the empirical strategy. Areas in gray are neither in treatment nor control as they overlap with the Millennium and Expo lines.

Figure 3: Effect of SkyTrain Expansion on Earnings



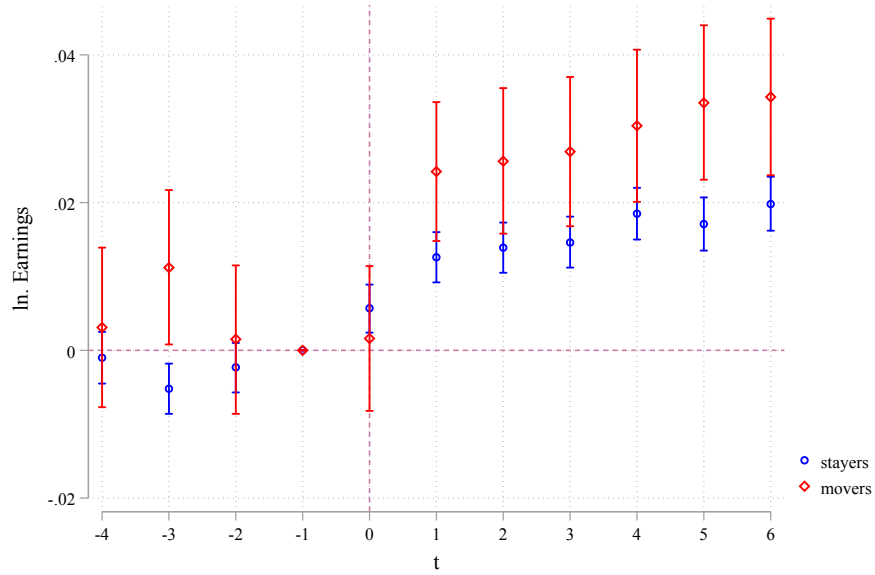
*Notes:* This figure plots estimates from equation (2) on an unbalanced sample of Vancouver residents. Time period is from 2005-2015, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

Figure 4: Effect of SkyTrain Expansion on Probability of Switching Employers



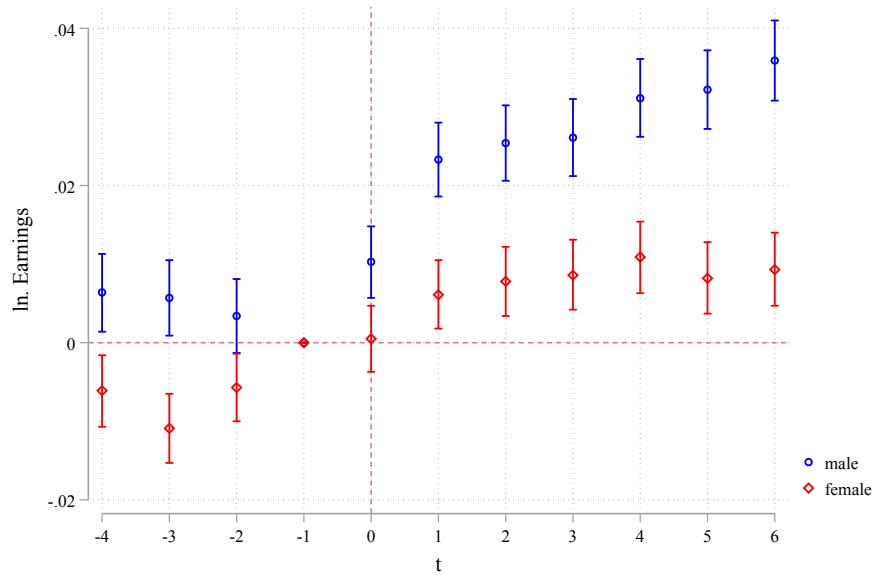
*Notes:* This figure plots yearly difference in probability of in switching 2008 primary employer between treatment and control groups. Specifically, plotted coefficients are estimates from the following regression  $D_{ikt} = \sum_{k=0}^6 \beta_k \mathbb{I}[k - 2009 = t] \times T_i + \alpha_{kt} + \epsilon_{ikt}$ , where  $D_{ikt}$  is a dummy indicating worker  $i$  has a new employer relative to 2008.

Figure 5: Effect of SkyTrain Expansion on Earnings By Job Switchers & Stayers



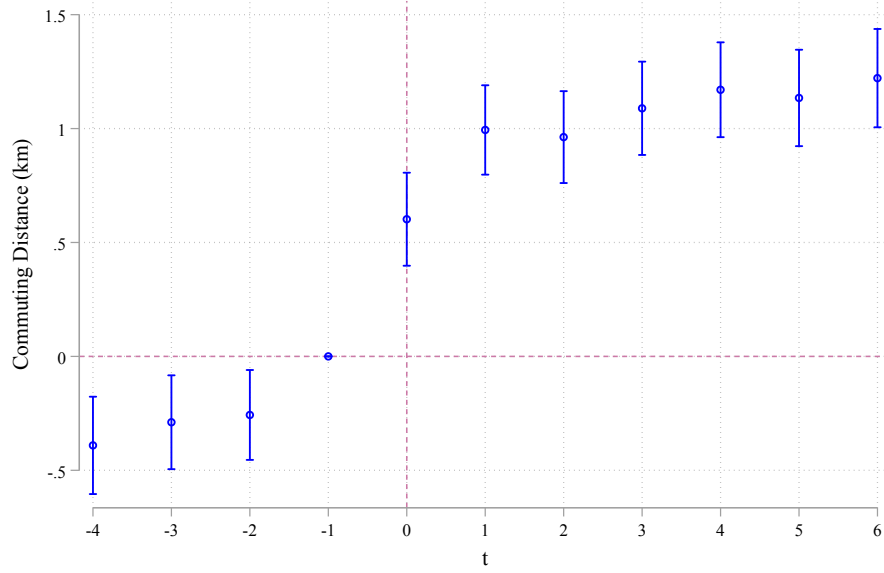
Notes: This figure plots estimates from (2) separately for job switchers and stayers. Estimation period is from 2005-2015, on an unbalanced sample of Vancouver residents, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

Figure 6: Effect of SkyTrain Expansion on Earnings By Gender



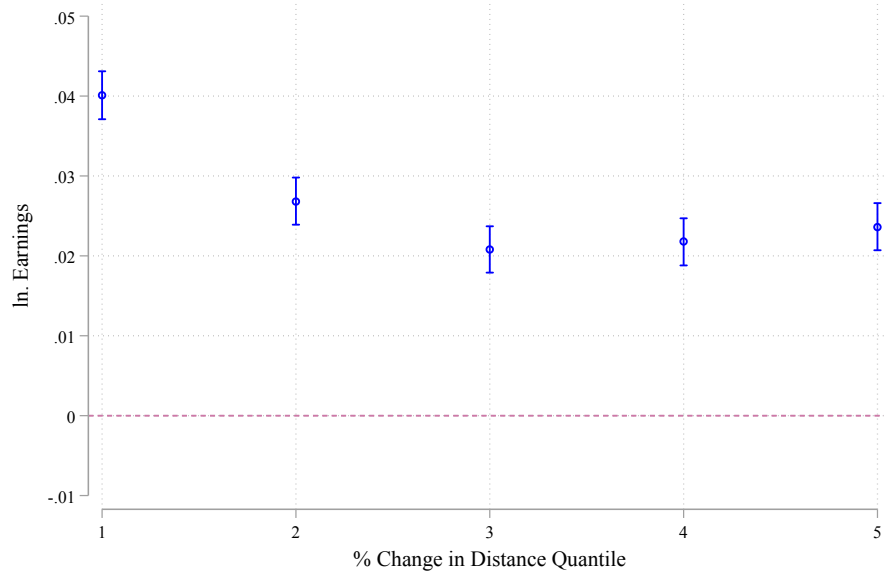
Notes: This figure plots estimates from (2) for those who had at least one job switch after the expansion, and the outcome variable is commuting distance measured as the euclidean distance between worker and firm location. Estimation period is from 2005-2015, on an unbalanced sample of Vancouver residents, and coefficients are estimated relative to period  $t = -1$ , one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

Figure 7: Effect of SkyTrain Expansion on Commuting Distance (km)



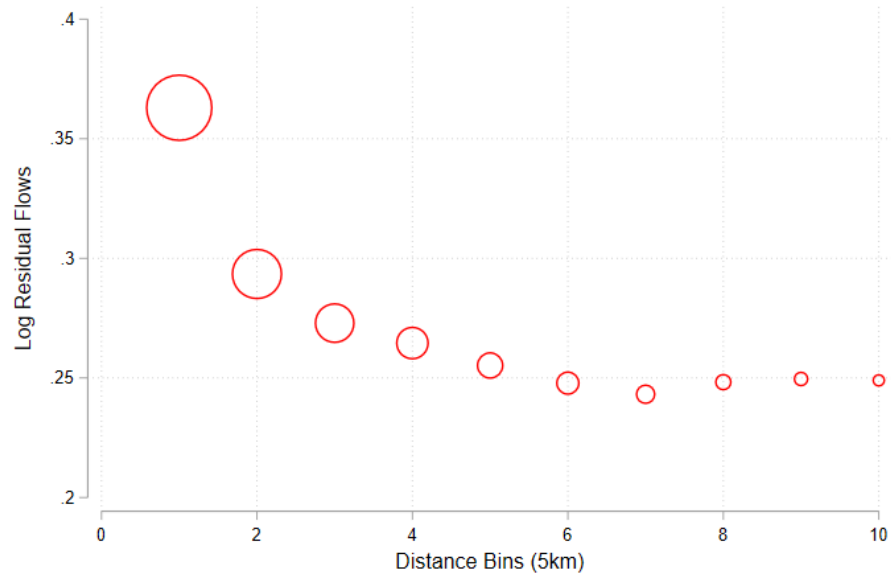
*Notes:* This figure plots estimates from (2), with the inclusion of firm  $\times$  year fixed effects to account for potential wage gains due to changes in firm productivity. Estimation period is from 2005-2015, on an unbalanced sample of Vancouver residents, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

Figure 8: Effect of SkyTrain Expansion on Annual Earnings By Treatment Intensity



*Notes:* This figure presents estimates from a static Diff-in-Diff specification with multiple treatment intensities. Treatment intensity is defined by the percentage change in distance to nearest station quantiles. The first quantile is the group that experienced the largest drop in distance (i.e. those with station very close to place of residence). The fifth quantile are workers with that had a positive, but low decrease in distance to nearest station. The control group are workers who had no change in distance to nearest station, and live more than 3km away from any subway station. Specification is  $Y_{it} = \alpha_i + \alpha_t + \sum_{d=1}^5 \beta_d \mathbb{I}[\% \Delta \in Q_d] \text{Post}_t \times \text{Treated}_i + \epsilon_{it}$  Where  $Q_d$  are the specified quantiles.

Figure 9: Residualized Number of Workers by Commuting Distance Bins



*Notes:* This figure plots the residualized number of workers at a given firm against 5 km travel distance bins. Residual flows are obtained from regressing the number of workers from neighborhood  $z$  of type  $k$  working at firm  $j$  on firm fixed effects using 2016 neighborhood-type-firm level data. I run the following  $\ln n_{zktj} = \alpha_{jt} + \epsilon_{zktj}$ . Note that firms hiring from only one neighborhood are dropped from this regression. However, zero hires are taken into account when using PPML to estimate equation (24).

## Tables

Table 1: Sample Summary Statistics 2001-2019

|                                        | All                | Male, 25-39 | Male 40-59 | Female, 25-39 | Female,40-59 |
|----------------------------------------|--------------------|-------------|------------|---------------|--------------|
| <b>Panel A: Workers</b>                |                    |             |            |               |              |
| Earnings                               | 67,000             | 64,000      | 84,000     | 49,000        | 57,000       |
| Age                                    | 42                 | 33          | 50         | 33            | 50           |
| Share Female                           | 0.36               | -           | -          | -             | -            |
| Share Ontario                          | 0.43               | 0.41        | 0.43       | 0.44          | 0.46         |
| Share Quebec                           | 0.25               | 0.25        | 0.26       | 0.24          | 0.25         |
| Share B.C.                             | 0.11               | 0.11        | 0.11       | 0.12          | 0.12         |
| Share Manuf.                           | 0.25               | 0.25        | 0.31       | 0.16          | 0.21         |
| Share Hospitality                      | 0.051              | 0.044       | 0.033      | 0.079         | 0.071        |
| Share Finance                          | 0.057              | 0.043       | 0.03       | 0.095         | 0.095        |
| Share Construction                     | 0.086              | 0.13        | 0.11       | 0.031         | 0.031        |
| Share Other Professional.              | 0.20               | 0.21        | 0.16       | 0.28          | 0.21         |
| Share commute $\leq 5$                 | 0.38               | 0.39        | 0.39       | 0.38          | 0.38         |
| Share commute $\in (5, 10]$            | 0.21               | 0.21        | 0.21       | 0.22          | 0.21         |
| Share commute $\in (10, 15]$           | 0.13               | 0.13        | 0.13       | 0.13          | 0.13         |
| Share commute $\in (15, 20]$           | 0.083              | 0.082       | 0.085      | 0.082         | 0.082        |
| Share commute $\in (20, 25]$           | 0.056              | 0.054       | 0.058      | 0.054         | 0.055        |
| Share commute $\in (25, 30]$           | 0.04               | 0.038       | 0.041      | 0.039         | 0.039        |
| Share commute $> 30$                   | 0.10               | 0.10        | 0.10       | 0.093         | 0.09         |
| <b>Panel B: Enterprises</b>            |                    |             |            |               |              |
| PD7 Number Of Employees                | 60                 |             |            |               |              |
| Share Multi establishment              | 0.10               |             |            |               |              |
| Number of $k$ types                    | 3.19               |             |            |               |              |
| Aggregate Shares $s_{kj}$              | $2 \times 10^{-6}$ |             |            |               |              |
| Neighborhood Specific Share $s_{z kj}$ | 0.00056            |             |            |               |              |
| Log. Revenue                           | 15.13              |             |            |               |              |
| Log. Value Added                       | 13.97              |             |            |               |              |
| Log. Total Payroll                     | 13.75              |             |            |               |              |
| Distance to Actual Location            | 23.25              |             |            |               |              |
| <b>Panel C: Neighborhoods</b>          |                    |             |            |               |              |
| Population                             | 10,000             |             |            |               |              |
| Rental Price Index                     | 1.1                |             |            |               |              |
| Log. Neighborhood Total Earnings       | 19.4               |             |            |               |              |
| Share Ontario                          | 0.41               |             |            |               |              |
| Share Quebec                           | 0.24               |             |            |               |              |
| Share British Columbia                 | 0.13               |             |            |               |              |
| Average within CMA share $\pi_{z k c}$ | 0.05               |             |            |               |              |
| No. Individual-year observations.      | 81,000,000         | 24,000,000  | 28,000,000 | 13,000,000    | 16,000,000   |
| No. Unique Individuals                 | 15,000,000         | 5,000,000   | 4,000,000  | 3,000,000     | 3,000,000    |
| No. Enterprise-year observations       | 2, 000,000         |             |            |               |              |
| No. Unique Enterprises                 | 316,000            |             |            |               |              |
| No. Unique Neighborhoods               | 1,034              |             |            |               |              |

Notes: This table presents summary statistics for the constructed sample used to estimate the model parameters. Panel A presents worker level summary statistics for all workers, and separately by worker types. Panel B includes Descriptives at the *enterprise* level. Panel C presents descriptives for the constructed *neighborhoods*. Observatio number are rounded to the nearest million in compliance with Statistics Canada confidentiality terms.



Table 2: Vancouver SkyTrain Expansion Summary Statistics by Treatment Status

| <b><i>Panel A: Pre-Expansion</i></b>  |         |           |
|---------------------------------------|---------|-----------|
|                                       | Control | Treatment |
| Age                                   | 42.85   | 42.09     |
| Share Female                          | 0.47    | 0.49      |
| Earnings                              | 68,000  | 78,000    |
| Log. Enterprise Total payroll         | 16.83   | 16.49     |
| Distance to nearest station 2008      | 12.38   | 6.94      |
| Commuting Distance                    | 15.55   | 15.26     |
| Res. Distance to CBD                  | 32.17   | 10.39     |
| Share Manuf.                          | 0.12    | 0.086     |
| Share Hospitality                     | 0.035   | 0.054     |
| Share Finance                         | 0.051   | 0.065     |
| Share Construction                    | 0.066   | 0.037     |
| Share Other Professional.             | 0.16    | 0.22      |
| No. Individual-year observations.     | 547,000 | 341,000   |
| No. Unique Individuals                | 165,000 | 109,000   |
| Average Change in Distance (%)        | 0       | -70       |
| <b><i>Panel B: Post-Expansion</i></b> |         |           |
|                                       | Control | Treatment |
| Age                                   | 45.95   | 44.68     |
| Share Female                          | 0.48    | 0.5       |
| Earnings                              | 70,000  | 79,000    |
| Log. Enterprise Total payroll         | 17.00   | 16.67     |
| Distance to nearest station 2008      | 12.29   | 6.96      |
| Commuting Distance                    | 15.71   | 14.45     |
| Res. Distance to CBD                  | 32.54   | 10.28     |
| Share Manuf.                          | 0.099   | 0.072     |
| Share Hospitality                     | 0.034   | 0.055     |
| Share Finance                         | 0.049   | 0.062     |
| Share Construction                    | 0.072   | 0.042     |
| Share Other Professional.             | 0.15    | 0.21      |
| No. Individual-year observations.     | 888,000 | 588,000   |
| No. Unique Individuals                | 170,000 | 116,000   |
| Average Change in Distance (%)        | 0       | -71       |

*Notes:* This table presents averages and shares pre and post SkyTrain expansion by treatment status for the estimation sample. Treatment is defined as whether or not an individual had *some* drop in distance to nearest station between 2008 and 2009. Panel A reports summary statistics prior to the expansion (2005-2008), and Panel B reports summary statistics post expansion (2009-2015). Res. distance to CBD refers to as the crow flies distances between Downtown Vancouver and a worker's place of residence in a given year. Earnings and log. Enterprise total payroll are in 2016 dollars. The category *other Professional* refers to workers in industries with 2-digit NAICS codes between 53-56. This includes Real Estate agents, managers, professional services, and administrative support. For confidentiality, number of observations and average earnings are rounded to the nearest thousand.

Table 3: Labor Supply Parameters by Group  $k$ 

| Labor Supply Parameters            |                | Male, 25-39 | Male, 40-60 | Female, 25-39 | Female, 40-60 |
|------------------------------------|----------------|-------------|-------------|---------------|---------------|
| Marginal Utility of Earnings       | $\beta_k$      | 2.818       | 1.182       | 3.229         | 1.354         |
| Within-Nest Independence Parameter | $\lambda_{kg}$ | 0.298       | 0.323       | 0.327         | 0.360         |
| Marginal Commuting Disutility      | $\eta_k$       | -0.00865    | -0.0223     | -0.00901      | -0.0265       |

Notes: This table presents labor supply parameters for each group  $k$ . Estimated  $\beta_k$  and  $\lambda_{kg}$  are estimated via 2SLS using the long-difference equation (22). Instruments used are one period changes in log revenue, one period changes in neighborhood-firm shares, and one period changes in establishment shares of other type of workers other than  $k$ . Equation (22) is estimated separately for each type  $k$ . The reported  $\lambda_{kg}$  are employment-weighted industry averages of all industry specific estimates from equation (22). The parameter  $\eta_k$  is estimated using 2016 data commuting flows between neighborhoods and firms. It is estimated using a commuting-flow weighted PPML regression separately for each  $(k, g)$  pair using equation (24). This results in 84 separate estimates. Each estimate is multiplied by the industry corresponding  $\frac{\lambda_{kg}}{\beta_k}$  to back out  $\eta_k$ . The reported parameter is the employment-weighted industry averages for each group  $k$ .

Table 4: Labor Supply Elasticities & Markdowns By Group  $k$ 

| <i>Panel A: All</i>           |                                                 | mean   | median | p10   | p90    |
|-------------------------------|-------------------------------------------------|--------|--------|-------|--------|
| Labor Supply Elasticity       | $\mathcal{E}_{kjt}$                             | 7.497  | 5.713  | 2.912 | 15.134 |
| Markdown                      | $\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt}+1}$ | 0.847  | 0.851  | 0.744 | 0.938  |
| <i>Panel B: Male, 25-39</i>   |                                                 | mean   | median | p10   | p90    |
| Labor Supply Elasticity       | $\mathcal{E}_{kjt}$                             | 10.749 | 11.687 | 6.517 | 15.047 |
| Markdown                      | $\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt}+1}$ | 0.907  | 0.921  | 0.867 | 0.938  |
| <i>Panel C: Male, 40-60</i>   |                                                 | mean   | median | p10   | p90    |
| Labor Supply Elasticity       | $\mathcal{E}_{kjt}$                             | 4.207  | 4.314  | 2.276 | 5.369  |
| Markdown                      | $\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt}+1}$ | 0.793  | 0.812  | 0.695 | 0.843  |
| <i>Panel D: Female, 25-39</i> |                                                 | mean   | median | p10   | p90    |
| Labor Supply Elasticity       | $\mathcal{E}_{kjt}$                             | 12.634 | 12.054 | 8.372 | 17.706 |
| Markdown                      | $\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt}+1}$ | 0.918  | 0.923  | 0.893 | 0.947  |
| <i>Panel E: Female, 40-60</i> |                                                 | mean   | median | p10   | p90    |
| Labor Supply Elasticity       | $\mathcal{E}_{kjt}$                             | 4.395  | 4.162  | 2.631 | 5.713  |
| Markdown                      | $\frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt}+1}$ | 0.795  | 0.806  | 0.725 | 0.851  |

Notes: This table presents elasticity and markdown distributions for all worker types, and each type  $k$  separately. Reported estimates are at the firm-level as defined by Equation (17), and equation (18) respectively. Mean elasticity and markdowns are *employment-weighted*. This is calculated for the period 2001-2019 and weights are at the  $(k, j, t)$  level; Median, p10, and p90 are unweighted establishment-type level distribution percentiles for the period 2001-2019.

Table 5: Labor Demand Parameter Estimates &amp; Distribution

| <i>Panel A: Distribution of Other Production Function Parameters</i> |                            | mean       | median | p10    | p90    |
|----------------------------------------------------------------------|----------------------------|------------|--------|--------|--------|
| Labor Productivity                                                   | $\gamma_{kjt}$             | 0.310      | 0.284  | 0.159  | 0.495  |
| Returns to Scale                                                     | $\alpha_{jt}$              | 0.289      | 0.240  | 0.076  | 0.578  |
| Total Factor Productivity                                            | $\ln(\tilde{\theta}_{jt})$ | 14.912     | 14.73  | 13.296 | 16.770 |
| <i>Panel B: Estimated Parameters From Equation (27)</i>              |                            | Est. Value |        |        |        |
| Labor Substitution Parameter                                         | $\rho$                     | 0.9103     |        |        |        |
| Persistence of Labor Productivity                                    | $\delta$                   | 0.734      |        |        |        |

*Notes:* This table reports the distribution of production function parameters from equation (16) at the  $(k, j, t)$  level for  $\gamma_{kjt}$  and at the  $(j, t)$  level for  $\alpha_{jt}$  and  $\ln(\tilde{\theta}_{jt})$ . The estimated parameters  $\rho$  and  $\delta$  are from estimating equation (27) using OLS. Given  $\rho$  other firm-specific production function parameters are recovered as outlined in the estimation procedure.

Table 6: Commuting &amp; Firm TFP

|                                       | $\ln(\tilde{\theta}_{jt})$ |                        |                       |                        |
|---------------------------------------|----------------------------|------------------------|-----------------------|------------------------|
|                                       | (1)                        | (2)                    | (3)                   | (4)                    |
| Commuting Distance (km)               | 0.011***<br>(0.00002)      | 0.0091***<br>(0.00002) | 0.009***<br>(0.00002) | 0.0077***<br>(0.00002) |
| Commuting Distance $\times$ Female    |                            | 0.0047***<br>(0.00004) |                       | 0.0045***<br>(0.00004) |
| Type $\times$ Year FEs                | Yes                        | Yes                    | No                    | No                     |
| Type $k \times$ CMA $\times$ Year FEs | No                         | No                     | Yes                   | Yes                    |
| Indv FEs                              | Yes                        | Yes                    | Yes                   | Yes                    |

*Notes:* This table presents estimated coefficients from regressing log TFP as estimated by  $\tilde{\theta}_{jt}$  on commuting distance in kilometers. Distance is measured as the euclidean distance between neighborhood centroids. All specifications include individual fixed effects. Column (1) includes worker  $k$  type by year fixed effects. Column (2) is similar but includes an interaction with a female dummy. Columns (3) and (4) include type  $k$  by year by CMA fixed effects.

Table 7: Model Simulation of SkyTrain Expansion

|                                                          | No Station | Millenium & Expo Line | Canada Line |
|----------------------------------------------------------|------------|-----------------------|-------------|
| <b>Wages, Elasticities &amp; Markdowns</b>               |            |                       |             |
| Average Wage                                             | -0.20 %    | -0.30%                | 1.0 %       |
| Average Elasticity $\mathcal{E}_{z kj}$                  | -0.13 %    | -0.11%                | -0.5 %      |
| Average Markdown                                         | -0.02 %    | -0.02%                | -0.08 %     |
| <b>Population &amp; Employment</b>                       |            |                       |             |
| Total Population                                         | -0.39%     | -0.14%                | 3.75%       |
| Total Employment                                         | -0.65%     | -0.30%                | 8.44%       |
| <b>Sorting</b>                                           |            |                       |             |
| Average Composite term                                   | -0.17%     | -0.34 %               | 1.7%        |
| Average $\tilde{\theta}_{jt}$                            | -0.08%     | -0.58%                | 5.12%       |
| <b>Welfare, Rents, Market Access &amp; Concentration</b> |            |                       |             |
| Average utility                                          | -0.11%     | -0.07%                | 0.45%       |
| Average Rents                                            | -0.70%     | -0.49%                | 8.50%       |
| Average $RMA_{zk}$                                       | -0.22%     | -0.13%                | 3.07%       |
| Average $GCI_{zk}$                                       | 0.57%      | 0.08%                 | -8.60 %     |

*Notes:* This table presents the percentage change between the baselin model in 2016 and the counterfactual economy in absence of the Canada Line. The no Canada Line economy is simulated by increasing commuting costs by a factor of 0.7 between neighborhoods on the Canada Line and other neighborhoods on the SkyTrain network. Column (1) corresponds to neighborhoods not on the SkyTrain network. Column (2) corresponds to neighborhoods near the Millenium & Expo lines, and column (3) correspnds to neighborhoods on the Canada Line. Labor market outcomes in wages, elasticites, markdowns, composite term, and  $\tilde{\theta}_{jt}$  are employment weighted averages. Rents,  $RMA_{zk}$  and  $GCI_{zk}$  are simple averages across  $(z, k)$  within each group.

Table 8: Model Simulation of Uniform Commuting Costs

| <b><i>Panel A: Within CMA</i></b> | $\% \Delta \mathbb{V}ar(w_{zk})$ | $\% \Delta \mathbb{V}ar(mdn_{zk})$ | $\% \Delta \mathbb{V}ar(mrpl_{zk})$ |
|-----------------------------------|----------------------------------|------------------------------------|-------------------------------------|
| Male, 25-39                       | -40.68                           | -15.23                             | -38.46                              |
| Male, 40-60                       | -42.34                           | -16.40                             | -37.13                              |
| Female, 25-39                     | -44.88                           | -12.24                             | -42.35                              |
| Female, 40-60                     | -52.53                           | -11.65                             | -46.47                              |
| <b><i>Panel B: Across CMA</i></b> | $\% \Delta \mathbb{V}ar(w_{zk})$ | $\% \Delta \mathbb{V}ar(mdn_{zk})$ | $\% \Delta \mathbb{V}ar(mrpl_{zk})$ |
| Male, 25-39                       | 2.68                             | -16.12                             | 1.37                                |
| Male, 40-60                       | -16.79                           | -14.07                             | -18.34                              |
| Female, 25-39                     | -20.50                           | -10.19                             | -19.38                              |
| Female, 40-60                     | -38.78                           | -10.65                             | -35.04                              |

*Notes:* This table presents percentage change in variance of mean wages, markdowns, and marginal product across neighborhoods. Panel A focuses on residual variance within a given Census Metropolitan Area, which is measured by regressing the underlying quantities on CMA fixed effects. Panel B considers differences across neighborhoods and CMAs. The percentage change compares the baseline model for 2016 with the counterfactual economy assuming all workers face the same commuting distance of 20 km to any employer within a 75 km radius of their chosen place of residence.

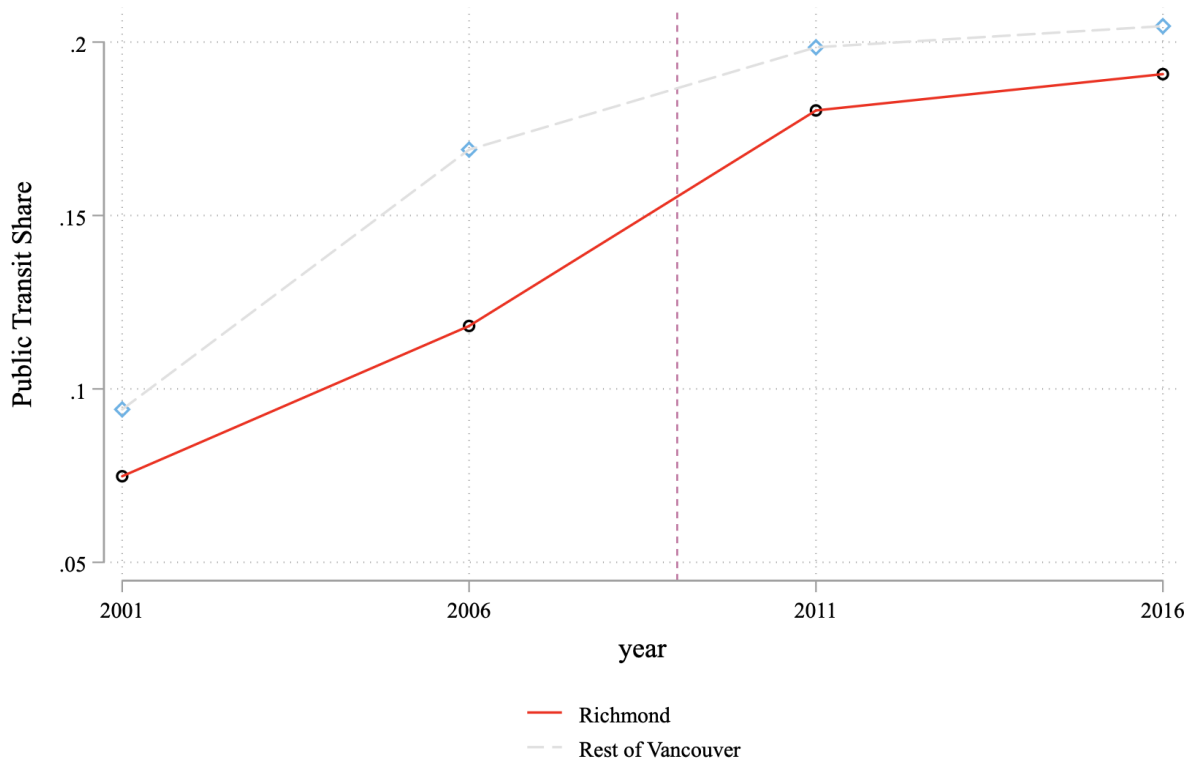
## Appendix A - Additional Figures

Figure A-1: Commuting Distribution for Canada & GTA municipalities

| CMA           | Municipality     | Employed Labourforce | % of Workers by Commuting Distance |             |               |               |               | Median Distance km | ST Mode% |
|---------------|------------------|----------------------|------------------------------------|-------------|---------------|---------------|---------------|--------------------|----------|
|               |                  |                      | Less than 5 km                     | 5 to 9.9 km | 10 to 14.9 km | 15 to 24.9 km | 25 km or more |                    |          |
| Barrie        | Barrie, CY       | 55,810               | 43%                                | 18%         | 3%            | 3%            | 33%           | 6                  | 10%      |
| Barrie        | Innisfil, T      | 12,555               | 11%                                | 8%          | 18%           | 13%           | 49%           | 24                 | 4%       |
| Barrie        | Springwater, TP  | 7,255                | 14%                                | 19%         | 14%           | 22%           | 31%           | 16                 | 4%       |
| Barrie, CMA   |                  | 75,625               | 35%                                | 17%         | 6%            | 7%            | 35%           | 9                  | 8%       |
| Hamilton      | Burlington, CY   | 74,340               | 32%                                | 18%         | 11%           | 14%           | 26%           | 10                 | 12%      |
| Hamilton      | Grimsby, T       | 10,440               | 24%                                | 8%          | 6%            | 26%           | 36%           | 21                 | 8%       |
| Hamilton      | Hamilton, C      | 207,120              | 33%                                | 25%         | 13%           | 12%           | 16%           | 8                  | 16%      |
| Hamilton, CMA |                  | 291,905              | 33%                                | 23%         | 12%           | 13%           | 19%           | 8                  | 15%      |
| Oshawa        | Clarington, MU   | 33,940               | 17%                                | 18%         | 14%           | 20%           | 31%           | 15                 | 6%       |
| Oshawa        | Oshawa, CY       | 59,660               | 39%                                | 22%         | 6%            | 8%            | 24%           | 7                  | 13%      |
| Oshawa        | Whitby, T        | 50,070               | 23%                                | 17%         | 9%            | 7%            | 44%           | 15                 | 13%      |
| Oshawa, CMA   |                  | 143,680              | 28%                                | 20%         | 9%            | 10%           | 33%           | 11                 | 12%      |
| Toronto       | Ajax, T          | 41,355               | 24%                                | 10%         | 5%            | 19%           | 42%           | 21                 | 17%      |
| Toronto       | Aurora, T        | 21,350               | 26%                                | 12%         | 5%            | 22%           | 35%           | 19                 | 12%      |
| Toronto       | Brampton, CY     | 188,995              | 21%                                | 24%         | 20%           | 20%           | 15%           | 11                 | 13%      |
| Toronto       | Caledon, T       | 24,990               | 17%                                | 7%          | 8%            | 29%           | 39%           | 21                 | 4%       |
| Toronto       | E.Gwillimbury, T | 9,410                | 16%                                | 19%         | 13%           | 10%           | 43%           | 16                 | 6%       |
| Toronto       | Georgina, T      | 17,595               | 17%                                | 4%          | 7%            | 21%           | 52%           | 27                 | 6%       |
| Toronto       | Halton Hills, T  | 25,290               | 26%                                | 6%          | 13%           | 28%           | 27%           | 17                 | 8%       |
| Toronto       | King, TP         | 8,100                | 10%                                | 11%         | 13%           | 27%           | 39%           | 21                 | 7%       |
| Toronto       | Markham, T       | 111,650              | 22%                                | 24%         | 17%           | 23%           | 14%           | 11                 | 17%      |
| Toronto       | Milton, T        | 26,255               | 26%                                | 5%          | 14%           | 26%           | 29%           | 17                 | 8%       |
| Toronto       | Mississauga, CY  | 294,340              | 24%                                | 25%         | 17%           | 20%           | 14%           | 10                 | 19%      |
| Toronto       | Mono, T          | 2,915                | 24%                                | 14%         | 8%            | 8%            | 46%           | 18                 | 4%       |
| Toronto       | New Tecumseth, T | 11,790               | 32%                                | 6%          | 7%            | 11%           | 45%           | 19                 | 9%       |
| Toronto       | Newmarket, T     | 34,655               | 41%                                | 9%          | 3%            | 11%           | 36%           | 11                 | 11%      |
| Toronto       | Oakville, T      | 71,450               | 27%                                | 15%         | 8%            | 17%           | 32%           | 15                 | 18%      |
| Toronto       | Orangeville, T   | 12,250               | 50%                                | 2%          | 1%            | 3%            | 44%           | 7                  | 9%       |
| Toronto       | Pickering, CY    | 39,755               | 20%                                | 10%         | 12%           | 26%           | 33%           | 18                 | 16%      |
| Toronto       | Richmond Hill, T | 68,820               | 19%                                | 19%         | 18%           | 27%           | 17%           | 13                 | 16%      |
| Toronto       | Toronto, C       | 1,016,825            | 34%                                | 28%         | 18%           | 14%           | 6%            | 8                  | 43%      |
| Toronto       | Uxbridge, TP     | 7,620                | 21%                                | 7%          | 5%            | 13%           | 54%           | 27                 | 7%       |
| Toronto       | Vaughan, CY      | 103,665              | 24%                                | 26%         | 16%           | 27%           | 8%            | 10                 | 11%      |
| Toronto       | W.Gwillimbury, T | 10,815               | 20%                                | 10%         | 19%           | 11%           | 40%           | 15                 | 7%       |
| Toronto       | W-Stouffville, T | 9,995                | 20%                                | 10%         | 11%           | 29%           | 30%           | 18                 | 8%       |
| Toronto, CMA  |                  | 2,160,020            | 29%                                | 23%         | 16%           | 18%           | 14%           | 9                  | 28%      |
|               | Canada           | 13,041,190           | 36%                                | 23%         | 13%           | 14%           | 14%           | 8                  | 19%      |

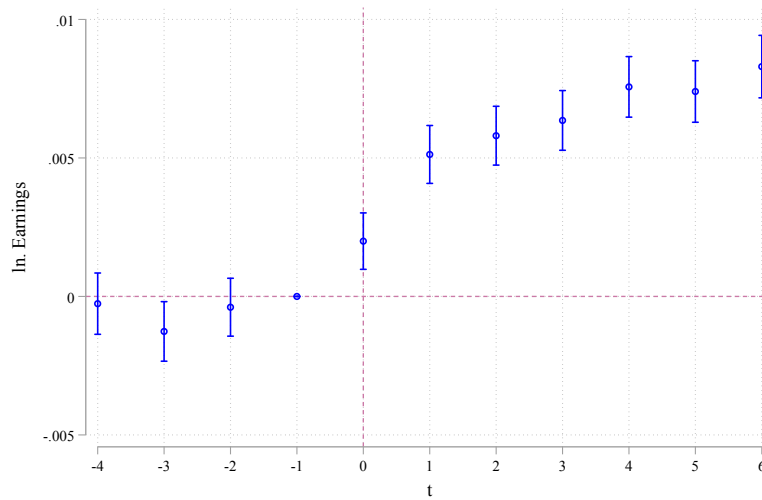
Source: 2006 Census

Figure A-2: Richmond & Rest of Metro Vancouver Public Transit Shares



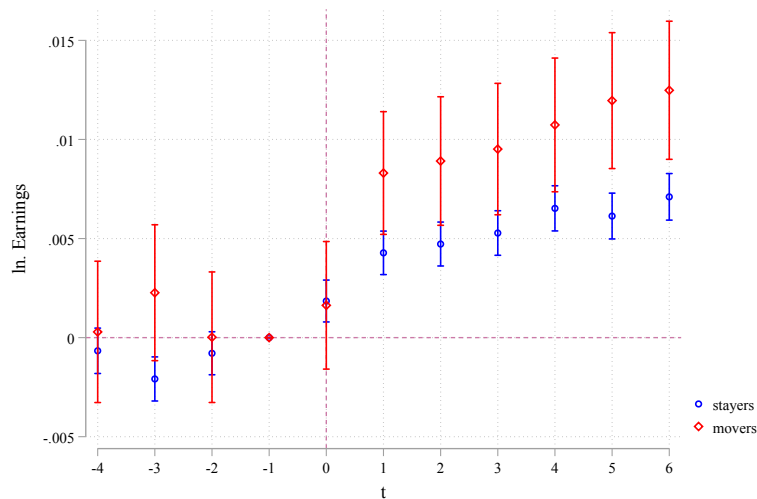
*Notes:* This Figure plots the share of public transit users in Richmond (red) and rest of Metro Vancouver area (light). Summary Statistics are from the long form 25% sample of the Canadian Census of Population for the corresponding years.

Figure A-3: Effect of SkyTrain Expansion on Earnings With Continuous Treatment



Notes: This figure plots estimates of the effect of going from p25 to p75 in the treatment exposure distribution after estimating the following TWFE specification:  $Y_{it} = \alpha_i + \alpha_t + \sum_{d=1}^5 \beta_d \% \Delta Dist Post_t + \epsilon_{it}$ . Time period is from 2005-2015, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

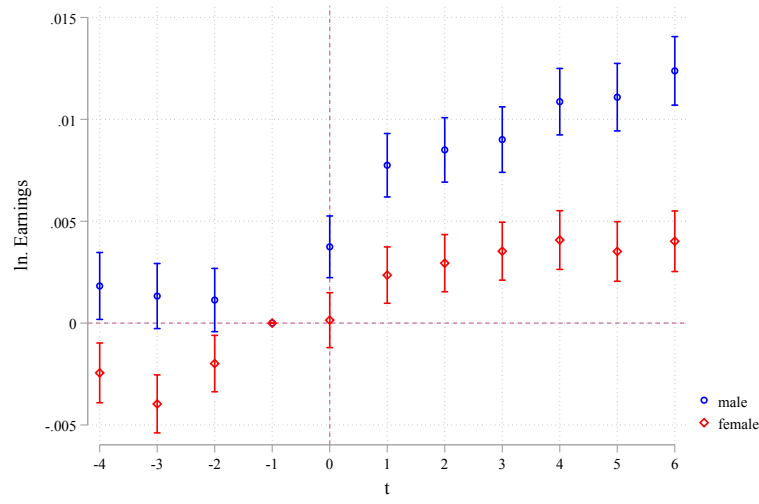
Figure A-4: Effect of SkyTrain Expansion on Earnings With Continuous Treatment By Job Switchers & Stayers



Notes: This figure plots estimates of the effect of going from p25 to p75 in the treatment exposure distribution after estimating the following TWFE specification:  $Y_{it} = \alpha_i + \alpha_t + \sum_{d=1}^5 \beta_d \% \Delta Dist Post_t + \epsilon_{it}$ . Time period is from 2005-2015, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.



Figure A-5: Effect of SkyTrain Expansion on Earnings With Continuous Treatment By Gender



*Notes:* This figure plots estimates of the effect of going from p25 to p75 in the treatment exposure distribution after estimating the following TWFE specification:  $Y_{it} = \alpha_i + \alpha_t + \sum_{d=1}^5 \beta_d \% \Delta Dist Post_t + \epsilon_{it}$ . Time period is from 2005-2015, and coefficients are estimated relative to period  $t = -1$  which corresponds to the year 2008, one year before the expansion. Specification includes individual fixed effects, and demographic group (gender  $\times$  age bin) by year fixed effects.

## Appendix B - Model Appendix

### B.1 Neighborhood $z$ Labor Supply to Firm $j$

Following the notation from (8). Recall that

$$s_{z kj} = \frac{\exp\left(\frac{\delta_{kj} + \mu_{z kj}}{\lambda_{kg}}\right) I_{z kg}^{\lambda_{kg} - 1}}{\sum_f I_{z kf}^{\lambda_{kf}}} \quad (\text{A.1})$$

The probability of remaining in non-employment is given by  $s_{zk0} = \frac{\exp(\delta_{k0})}{\sum_f I_{z kf}^{\lambda_{kf}}}$

$$\begin{aligned} \Rightarrow \frac{s_{z kj}}{s_{zk0}} &= \frac{\exp\left(\frac{\delta_{kj} + \mu_{z kj}}{\lambda_{kg}}\right) I_{z kg}^{\lambda_{kg}}}{\exp(\delta_{k0}) I_{z kg}} = \frac{s_{z kj|g} I_{z kg}^{\lambda_{kg}}}{\exp(\delta_{k0})} = s_{z kj|g} \frac{\exp(\delta_{kj} + \mu_{z kj})}{\exp(\delta_{k0})} \left[ \frac{I_{z kg}}{\exp\left(\frac{\delta_{kj} + \mu_{z kj}}{\lambda_{kg}}\right)} \right]^{\lambda_{kg}} \\ &= \frac{\exp(\delta_{kj} + \mu_{z kj})}{\exp(\delta_{k0})} s_{z kj|g}^{1 - \lambda_{kg}} \\ &= \frac{w_{kj}^{\beta_k} u_{kj} \exp(\eta_k d_{z,z(j)})}{w_{k0}^{\beta_k}} s_{z kj|g}^{1 - \lambda_{kg}} \\ \Rightarrow \ln \frac{s_{z kjt}}{s_{zk0t}} &= \beta_k \ln \frac{w_{kjt}}{w_{k0t}} + (1 - \lambda_{kg}) \ln s_{z kj|gt} - \tilde{\eta}_k d_{z,z(j)} + \ln u_{z kjt} \end{aligned}$$

The second equality follows from noting that the probability of choosing firm  $j$  conditional on choosing nest  $N_g$  is given by

$$s_{z kj|g} = \frac{\exp\left(\frac{\delta_{kj} + \mu_{z kj}}{\lambda_{kg}}\right)}{\underbrace{\sum_{j' \in N_g} \exp\left(\frac{\delta_{kj'} + \mu_{z kj'}}{\lambda_{kg}}\right)}_{I_{z kg}}}$$

### B.2 Neighborhood $z$ Labor Supply *Elasticity* to Firm $j$

Labor supply elasticity between a given neighborhood firm pair depends on the sum of elasticity conditional on workers' spatial allocation with population elasticity of the neighborhood with respect to the firm's wages.

**Elasticity Conditional on Workers' Spatial Allocation** This section derives equation (9) from the model. This is the firm labor supply elasticity conditional on workers' spatial distribution.

First note that the share of workers from neighborhood  $z$  working for firm  $j$  is the probability of

working in nest  $g$  where  $j \in N_g$  and the conditional probability of choosing this firm. We can rewrite (8) as the following

$$s_{z kj} = \underbrace{\frac{\exp\left(\frac{\delta_{kj} + \mu_{z kj}}{\lambda_{kg}}\right)}{I_{z kg}}}_{s_{z kj|g}} \times \frac{\overbrace{I_{z kg}^{\lambda_{kg}}}}{s_{z kg}} \quad (\text{A.2})$$

$$= s_{z kj|g} s_{z kg} \quad (\text{A.3})$$

Where  $I_{z kf} := \sum_{l \in N_f} \exp\left\{\frac{\delta_{kl} + \mu_{z kl}}{\lambda_{kf}}\right\}$ . Taking logs on both sides of the above equation we get

$$\ln s_{z kj} = \frac{\beta_k}{\lambda_{kg}} \ln w_{kj} + \frac{1}{\lambda_{kg}} \ln u_{kj} - \frac{\eta_k}{\lambda_{kg}} d_{z, z(j)} + (\lambda_{kg} - 1) \ln I_{z kg} - \ln \left( \sum_f I_{z kf}^{\lambda_{kf}} \right)$$

Assuming firms behavior affects the wage index in their local labor market, and also globally means that

$$\begin{aligned} \frac{\partial \ln I_{z kg}}{\partial \ln w_{kj}} &\neq 0 \\ \frac{\partial \ln \left( \sum_f I_{z kf}^{\lambda_{kf}} \right)}{\partial \ln w_{kj}} &\neq 0 \end{aligned}$$

Deriving the first of these two equations we have

$$\begin{aligned} \frac{\partial \ln I_{z kg}}{\partial \ln w_{kj}} &= \frac{\partial \ln \left( \sum_{j' \in N_g} w_{kj'}^{\frac{\beta_k}{\lambda_{kg}}} u_{kj'}^{\frac{1}{\lambda_{kg}}} \exp(-\eta_k d_{z, j'}) \right)}{\partial w_{kj}} w_{kj} \\ &= \left[ \frac{\frac{\beta_k}{\lambda_{kg}} w_{kj}^{\frac{\beta_k}{\lambda_{kg}}} u_{kj}^{\frac{1}{\lambda_{kg}}} \exp(-\frac{\eta_k}{\lambda_{kg}} d_{z, j})}{w_{kj} \sum_{j' \in N_g} w_{kj'}^{\frac{\beta_k}{\lambda_{kg}}} u_{kj'}^{\frac{1}{\lambda_{kg}}} \exp(-\frac{\eta_k}{\lambda_{kg}} d_{z, j'})} \right] w_{kj} \\ &= \frac{\beta_k}{\lambda_{kg}} s_{z kj|g} \end{aligned} \quad (\text{A.4})$$

The last line follows from (A.2). Deriving the second of these equations follows a similar line of reasoning

$$\begin{aligned}
\frac{\partial \ln(\sum_f I_{z kf}^{\lambda_{kf}})}{\partial \ln w_{kj}} &= \frac{\partial \ln(\sum_f I_{z kf}^{\lambda_{kf}})}{\partial w_{kj}} w_{kj} \\
&= \frac{\partial I_{z kg}^{\lambda_{kg}}}{\partial w_{kj}} \frac{w_{kj}}{\sum_f I_{z kf}^{\lambda_{kf}}} \\
&= \lambda_{kg} I_{z kg}^{\lambda_{kg}-1} \frac{\partial I_{z kg}}{\partial w_{kj}} \frac{w_{kj}}{\sum_f I_{z kf}^{\lambda_{kf}}} \\
&= \lambda_{kg} I_{z kg}^{\lambda_{kg}-1} \times I_{z kg} \times \frac{\partial \ln I_{z kg}}{\partial \ln w_{kj}} \times \frac{1}{\sum_f I_{z kf}^{\lambda_{kf}}} \\
&= \lambda_{kg} \frac{\beta_k}{\lambda_{kg}} s_{z kg} s_{z kj|g} \\
&= \beta_k s_{z kj}
\end{aligned} \tag{A.5}$$

The last two equalities follow from the previous derivation and (A.2). Putting these together we get that

$$\begin{aligned}
\frac{\partial \ln s_{z kj}}{\partial \ln w_{kj}} &= \underbrace{\frac{\beta_k}{\lambda_{kg}}}_{\text{monopsonistic (LMS)}} + \underbrace{(\lambda_{kg} - 1) \frac{\ln I_{z kg}}{\ln w_{kj}}}_{\text{oligopsonistic (BHM)}} - \underbrace{\frac{\ln(\sum_f I_{z kf}^{\lambda_{kf}})}{\ln w_{kj}}}_{\text{aggregate competition (CKMM)}} \\
&= \frac{\beta_k}{\lambda_{kg}} + (\lambda_{kg} - 1) \frac{\beta_k}{\lambda_{kg}} s_{z kj|g} - \beta_k s_{z kj} \\
&= \beta_k \left[ \frac{1}{\lambda_{kg}} + \left(1 - \frac{1}{\lambda_{kg}}\right) s_{z kj|g} - s_{z kj} \right] \\
&= \mathcal{E}_{z kj}
\end{aligned} \tag{A.6}$$

**Elasticity of Neighborhood  $z$  population to Firm  $j$**  Since firms' are non-atomisitic and can set wages to influence worker's residential choice, the labor supply elasticity facing a firm is the sum of population elasticity, and labor supply elasticity conditional on location. Following the same argument as above, and taking into account the distributional assumption in  $v_{it}$  in (4) we can show that

$$\zeta_{z kjt} = \varphi \beta_k s_{z kjt} \left[ \frac{1}{\sigma} + \left(1 - \frac{1}{\sigma}\right) \pi_{z k|ct} - \pi_{z kct} \right] \tag{A.7}$$

### B.3 Aggregate Labor Supply to Firm $j$

In this section I show how to aggregate up to get the firm-specific, neighborhood invariant, labor supply and its elasticity for a given type. I show that the latter is an average of the elasticities from all neighborhoods.

#### Aggregation

$$\begin{aligned}\ell_{kj} &= \sum_{z \in \mathcal{Z}} s_{z kj} \ell_{zk} \\ &= w_{kj}^{\frac{\beta_k}{\lambda_{kg}}} u_{kj}^{\frac{1}{\lambda_{kg}}} \sum_{z \in \mathcal{Z}} \exp\left(-\frac{\eta_k}{\lambda_{kg}} d_{z,o(j)}\right) \frac{I_{zkg}^{\lambda_{kg}-1}}{(\sum_f I_{z kf}^{\lambda_{kf}})} \ell_{zk}\end{aligned}$$

Labor supply to firm  $j$  is inversely proportional to a population weighted average of distance between the firm from and potential workforce. This indicates that businesses that are located further away from their residential locations, e.g. those in different neighborhoods, stand to experience a larger labor supply shift as a result of remote work introduction.

**Elasticity  $\mathcal{E}_{kj}$**  First, let  $\mu_{zk}$  be the share of workers of type  $k$  who reside in neighborhood  $z$ . Then firm specific labor supply is given by

$$s_{kj} = \sum_{z \in \mathcal{Z}} s_{z kj} \mu_{zk}$$

Leveraging the linearity of the summation operator we have

$$\begin{aligned}\mathcal{E}_{kj} &:= \frac{\partial s_{kj}}{\partial w_{kj}} \frac{w_{kj}}{s_{kj}} = \sum_{z \in \mathcal{Z}} \frac{\partial s_{z kj}}{\partial w_{kj}} \frac{w_{kj}}{s_{kj}} \mu_{zk} \\ &= \sum_{z \in \mathcal{Z}} \frac{\partial s_{z kj}}{\partial w_{kj}} \frac{w_{kj}}{s_{z kj}} \frac{s_{z kj}}{s_{kj}} \mu_{zk} \\ &= \sum_{z \in \mathcal{Z}} \frac{\partial \ln s_{z kj}}{\partial \ln w_{kj}} \frac{s_{z kj}}{s_{kj}} \mu_{zk} \\ &= \sum_{z \in \mathcal{Z}} (\mathcal{E}_{z kj} + \underbrace{\zeta_{z kj t}}_{\omega_{z kj}}) \frac{s_{z kj}}{s_{kj}} \mu_{zk} \\ &= \sum_{z \in \mathcal{Z}} (\mathcal{E}_{z kj} + \zeta_{z kj t}) \omega_{z kj}\end{aligned}$$

The third equality follows from multiplying and dividing each term in the summation by the respective  $s_{z kj}$ . Recall that

$$\omega_{z kj} = \frac{\mathbb{P}(z, j|k)}{\mathbb{P}(j|k)} = \frac{\# \text{ workers } (z, k, j)}{\# \text{ workers } (k, j)}$$

This proves that firm specific labor supply of type  $k$  is a weighted average of the neighborhood to firm specific labor supplies. The weights take into account that the sum of log is not equal to log of sums.

#### B.4 Model Implied Gravity Equation

$$\ln \left( \frac{s_{z kjt}}{s_{z kj0t}} \right) = \beta_k \ln \left( \frac{w_{k jt}}{w_{k 0t}} \right) + (1 - \lambda_{kg}) \ln (s_{z kj|gt}) - \tilde{\eta}_k d_{zo(j)} + \ln u_{z kjt}$$

**Assumption 5.** Location-type-firm specific amenities take have the following separable form

$$u_{z kjt} = u_{k jt} \times u_{z kt} \times \epsilon_{z kjt}$$

and the residual error term satis

$$\mathbb{E}[\epsilon_{z kjt}|z, j, k, t] = 1$$

The second part of this assumption allows the estimation of  $\eta_k$  using PPML as shown by shown [Silva and Tenreiro \(2006\)](#). Equation (A.1) combined with the separability assumption implies

$$\begin{aligned} n_{z kjt} &= \exp \left( \frac{\eta_k d_{zj}}{\lambda_{kg}} \right) \exp \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{z kjt}}{\lambda_{kg}} \right) \frac{I_{z kg}^{\lambda_{kg}}}{\sum_f I_{z kf}^{\lambda_{kf}}} L_{zk} \\ &= \epsilon_{z kjt} \exp \left( \frac{\eta_k d_{zj}}{\lambda_{kg}} \right) \exp \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj}}{\lambda_{kg}} \right) \frac{\exp \left( \frac{\ln u_{zk}}{\lambda_{kg}} \right) \tilde{I}_{z kg}^{\lambda_{kg}}}{\sum_f \tilde{I}_{z kf}^{\lambda_{kf}}} L_{zk} \\ &= \epsilon_{z kjt} \times \exp \left( \frac{\eta_k d_{zj}}{\lambda_{kg}} \right) \times \exp (\ln FE_{kj}) \times \exp (\ln FE_{zk}) \\ &= \epsilon_{z kjt} \times \exp \left( \frac{\eta_k d_{zj}}{\lambda_{kg}} + \ln FE_{kj} + \ln FE_{zk} \right) \end{aligned}$$

$$\begin{aligned} FE_{kj} &= \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj}}{\lambda_{kg}} \\ \Rightarrow \ln u_{kj} &= \lambda_{kg} FE_{kj} - \beta_k \ln \frac{w_{kj}}{w_{k0}} \end{aligned}$$

Given  $\ln u_{kj}$  we can use the log odds of employment relative to non-employment to back out  $\ln u_{zkt}$  in the following manner. First, define  $s_{zk0}$  to be the probability of employment at any firm. Given the separability assumption, this could be written as

$$\begin{aligned}
I_{z kf} &= \exp \left( \frac{\ln u_{zk}}{\lambda_{kf}} \sum_{j \in N_f} \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj} + d_{zj}}{\lambda_{kg}} \right) \right)^{\lambda_{kf}} \\
&= u_{zk} \exp \left[ \sum_{j \in N_f} \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj} + d_{zj}}{\lambda_{kg}} \right) \right]^{\lambda_{kf}} \\
s_{zk0} &= \frac{u_{zk} \sum_f \exp \left[ \sum_{j \in N_f} \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj} + d_{zj}}{\lambda_{kg}} \right) \right]^{\lambda_{kf}}}{1 + \sum_f I_{z kf}^{\lambda_{kf}}} \\
\frac{s_{zk0}}{s_{zk0}} &= u_{zk} \sum_f \exp \left[ \sum_{j \in N_f} \left( \frac{\beta_k \ln \frac{w_{kj}}{w_{k0}} + \ln u_{kj} + d_{zj}}{\lambda_{kg}} \right) \right]^{\lambda_{kf}}
\end{aligned}$$

## B.5 Equilibrium Uniqueness

The goal is to derive the firm-level super-elasticity with respect to own wages and other firms' wages. I will show that it is a composite of the average super elasticity as well as the within firm variance in elasticities. The same applies to the cross super elasticity.

### Useful Properties

1. Define  $\mu_{zk}$  to be the share of workers of type  $k$  living in neighborhood  $z$
2. Recall that the firm level elasticity was the weighted average of elasticities across all neighborhoods

$$\mathcal{E}_{kj} = \sum_{z \in \mathcal{Z}} \mathcal{E}_{z kj} \omega_{z kj}$$

where

$$\omega_{z kj} = \frac{\mathbb{P}(z, j|k)}{\mathbb{P}(j|k)} = \frac{\# \text{ workers } (z, k, j)}{\# \text{ workers } (k, j)} = \frac{s_{z kj} \mu_{zk}}{\sum_{z'} s_{z' kj} \mu_{z' k}} = \mathbb{P}(z|j, k)$$

3. So we have that

$$\mathcal{E}_{kj} = \sum_{z \in \mathcal{Z}} \mathcal{E}_{z kj} \frac{s_{z kj} \mu_{zk}}{s_{kj}}$$

4. Recall too that

$$\mathcal{E}_{z kj} = \beta_k \left[ \frac{1}{\lambda_{kg}} + \left( 1 - \frac{1}{\lambda_{kg}} \right) s_{z kj|g} - s_{z kj} \right]$$

5. Combining 2 and 3 implies the firm level productivity could be written as the following:

6.

$$\mathcal{E}_{kj} = \beta_k \left[ \frac{1}{\lambda_{kg}} \left( 1 - \frac{1}{\lambda_{kg}} \right) \tilde{s}_{z kj|g} - \tilde{s}_{z kj} \right] := \mathbb{E}[\mathcal{E}_{z kj}|j, k]$$

where

$$\tilde{x} = \sum_z x \mathbb{P}(z|j, k) =: \mathbb{E}[x|j, k]$$

Equipped with this property I will now derive the semi super elasticity

$$\begin{aligned} \frac{\partial \mathcal{E}_{kj}}{\partial \ln w_{kj}} &= \sum \frac{\partial}{\partial \ln w_{kj}} \left( \frac{\mathcal{E}_{z kj} s_{z kj} \mu_{zk}}{s_{kj}} \right) \\ &= s_{kj}^{-1} \left[ \sum_z \frac{\partial \mathcal{E}_{z kj}}{\partial \ln w_{kj}} s_{z kj} \mu_{zk} + \sum_z \frac{\partial s_{z kj}}{\partial \ln w_{kj}} \mathcal{E}_{z kj} \mu_{zk} \right] - \frac{\partial s_{kj}}{\partial \ln w_{kj}} s_{kj}^{-2} \sum_{z \in \mathcal{Z}} \mathcal{E}_{z kj} s_{z kj} \mu_{zk} \\ &= s_{kj}^{-1} \left[ \sum_z \frac{\partial \mathcal{E}_{z kj}}{\partial \ln w_{kj}} s_{z kj} \mu_{zk} + \sum_z \frac{\partial s_{z kj}}{\partial \ln w_{kj}} \mathcal{E}_{z kj} \mu_{zk} \right] - \mathcal{E}_{kj}^2 \\ &= \sum_z \frac{\partial \mathcal{E}_{z kj}}{\partial \ln w_{kj}} \frac{s_{z kj} \mu_{zk}}{s_{kj}} + \sum_z \left( \mathcal{E}_{z kj}^2 \frac{s_{z kj} \mu_{zk}}{s_{kj}} \right) - \mathcal{E}_{kj}^2 \\ &= \underbrace{\mathbb{E} \left[ \frac{\partial \mathcal{E}_{z kj}}{\partial \ln w_{kj}} \middle| j, k \right]}_{<0} + \underbrace{\mathbb{V}ar \left[ \mathcal{E}_{z kj} \middle| j, k \right]}_{>0} \end{aligned}$$

The third equality follows from noting that

$$\frac{\partial s_{kj}}{\partial \ln w_{kj}} s_{kj}^{-1}$$

And

$$s_{kj}^{-1} \sum_{z \in \mathcal{Z}} \mathcal{E}_{z kj} s_{z kj} \mu_{zk}$$

are two different ways of writing down the firm level elasticity term. The fourth equality follows from multiplying and dividing the second term in the square bracket by  $s_{z kj}$ . The fifth equality follows from the definition of conditional expectation and variance.



The neighborhood level superelasticities are negative as in a standard nested logit framework. This could be viewed as the intensive margin of elasticity adjustment. However, since firms could source labor from any neighborhood the within-firm distribution of elasticities also matters, as presented by the second moment. This is the extensive margin, as firms could hire workers from further away neighborhoods as they increase the offered wages. When firms offer higher wages there are two effects. One, workers from a given neighborhood become more inelastic as the firm grows and increases its market share. This allows the firm to re-adjust and offer lower wages until converging to some middle point. Two, when a demand shock hits the firm it changes the workforce compositions. So that when firms offer higher wages it attracts workers from further away neighborhood who are more sensitive to wages changes due to commuting. This increases firm level elasticity and reduces the firm's wage setting power. By this process wage changes at a firm can become self reinforcing and can result in multiple equilibria. The question is now: *What are the restrictions that would allow there to be a unique equilibrium? I will now derive the variance term in the above equation more explicitly.*

$$\begin{aligned}
\mathbb{V}ar[\mathcal{E}_{zjk}|j, k] &= \sum_z (\mathcal{E}_{zjk} - \mathcal{E}_{kj})^2 \frac{s_{zjk} \mu_{zk}}{s_{kj}} \\
&= \sum_z \left[ \left(1 - \frac{1}{\lambda_{kg}}\right) (s_{zjk} - \tilde{s}_{kj}) - (s_{zjk} - \tilde{s}_{zjk}) \right]^2 \frac{s_{zjk} \mu_{zk}}{s_{kj}} \\
&= \left(1 - \frac{1}{\lambda_{kg}}\right)^2 \sum_z (s_{zjk} - \tilde{s}_{kj})^2 \frac{s_{zjk} \mu_{zk}}{s_{kj}} + \sum_z (s_{zjk} - \tilde{s}_{zjk})^2 \frac{s_{zjk} \mu_{zk}}{s_{kj}} \\
&\quad + 2 \left(1 - \frac{1}{\lambda_{kg}}\right) \sum_z (s_{zjk} - \tilde{s}_{kj}) (s_{zjk} - \tilde{s}_{zjk}) \frac{s_{zjk} \mu_{zk}}{s_{kj}} \\
&= \left(1 - \frac{1}{\lambda_{kg}}\right)^2 \mathbb{V}ar(s_{zjk}|j, k) + \mathbb{V}ar(s_{zjk}|j, k) - 2 \left(1 - \frac{1}{\lambda_{kg}}\right) Cov(s_{zjk}, s_{zjk}|j, k)
\end{aligned}$$

The second equality uses the alternative definition of firm level elasticity in bullet point (5) from the useful properties section. The rest follows the definitions of variance and covariance. This could be further simplified by noting the following

$$\begin{aligned}
\mathbb{V}ar(s_{zjk}|j, k) &= \mathbb{V}ar(s_{zjk} s_{zkg}|j, k) \\
&= \mathbb{V}ar(s_{zjk}|j, k)
\end{aligned}$$

Conditional on being at firm  $j$  the probability of working in industry  $g$  one. As the industry of the

firm is implicitly being conditioned on. The same rationale also implies the following:

$$Cov(s_{zkg|g}, s_{zkg}|j, k) = Cov(s_{zkg|g}, s_{zkg|g}s_{zkg}|j, k) = Var(s_{zkg|g}|j, k)$$

So the last equation from the previous derivations implies the following:

$$Var[\mathcal{E}_{zkg}|j, k] = \frac{1}{\lambda_{kg}^2} Var(s_{zkg|g}|j, k) \quad (\text{A.8})$$

And the super firm level labor supply elasticity could then be written as

$$\frac{\partial \mathcal{E}_{kj}}{\partial \ln w_{kj}} = \underbrace{\mathbb{E}\left[\frac{\partial \mathcal{E}_{zkg}}{\partial \ln w_{kj}} \middle| j, k\right]}_{\text{intensive margin}} + \underbrace{\frac{1}{\lambda_{kg}^2} Var(s_{zkg|g}|j, k)}_{\text{extensive margin}} \quad (\text{A.9})$$

## Appendix C - Simulation Algorithm

I simulate the model following the Jacobi algorithm over wages and rental prices. The algorithm makes use of two primary equations. One, firm best response function, and housing market clearing condition.

Firm best response functions

$$w_{kjt} = \frac{\mathcal{E}_{kjt}}{\mathcal{E}_{kjt} + 1} P_{jt} \theta_{jt} \sum_{k \in \mathcal{K}} \left( \gamma_{kjt} \ell_{kjt}^\rho \right)^{\frac{\alpha}{\rho} - 1} \quad (\text{A.10})$$

Housing Market Clearing:

$$r_{zt} = \frac{\gamma}{H_{zt}^s} \underbrace{\sum_{k \in \mathcal{K}} m_{kt} \mu_{zkt} \sum_{j \in \mathcal{J}} w_{kjt} s_{zkt}}_{\text{Total Earnings in neighborhood } z := E_{zt}} \quad (\text{A.11})$$

The housing clearing follows from Equation (14), and setting  $H_{zt}^d(r_{zt}) = H_{zt}^s$

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### Algorithm 1: Under-relaxed Jacobi Algorithm

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- 1: Initialize vector of wages  $\mathbf{w}_{\mathbf{kj}}^0$  for all  $\mathcal{K} \times \mathcal{J}$  and rents  $\mathbf{r}_z^0$  for all locations  $z \in \mathcal{Z}$
  - 2: **repeat**
  - 3:   Use equations (10) and (12) to generate neighborhood shares  $\mu_{zk}$
  - 4:   Given  $\mu_{zk}$ , use equation (8) to generate  $s_{zkt}$
  - 5:   Given  $s_{zkt}$ , use equation (9) to generate  $\mathcal{E}_{zkt}$ . And equation (13) to generate  $\zeta_{zkt}$
  - 6:   Given  $s_{zkt}, \mu_{zk}, \mathcal{E}_{zkt}, \zeta_{zkt}$  use equation (17) to generate  $\mathcal{E}_{kj}$
  - 7:   Given  $s_{zkt}, \mu_{zk}$  sum to get  $l_{kj} = \sum_z s_{zkt} \mu_{zkt} \ell_k$  and calculate  $P_{jt} \theta_{jt} \sum_{k \in \mathcal{K}} \left( \gamma_{kjt} \ell_{kjt}^\rho \right)^{\frac{\alpha}{\rho} - 1}$
  - 8:   Given  $s_{zkt}, \mu_{zk}, \mathbf{w}_{\mathbf{kj}}^{new}$  construct  $E_{zt}$  in equation (A.11)
  - 9:   Use equation (A.10) to define  $\mathbf{w}_{\mathbf{kj}}^{new}$ , and equation (A.11) to define  $\mathbf{r}_z^{new}$
  - 10:   Update wages using  $\mathbf{w}_{\mathbf{kj}}^{n+1} = \lambda \mathbf{w}_{\mathbf{kj}}^{new} + (1 - \lambda) \mathbf{w}_{\mathbf{kj}}^n$
  - 11:   Update rents using  $\mathbf{r}_z^{n+1} = \lambda \mathbf{r}_z^{new} + (1 - \lambda) \mathbf{r}_z^n$
  - 12: **until**  $\|\mathbf{w}_{\mathbf{kj}}^{n+1} - \mathbf{w}_{\mathbf{kj}}^n\|_{max} < \epsilon$  and  $\|\mathbf{r}_z^{n+1} - \mathbf{r}_z^n\|_{max} < \epsilon$
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